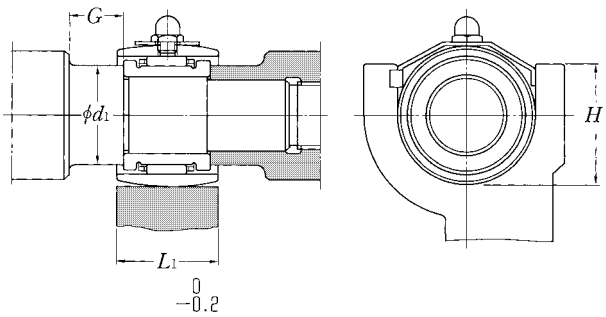
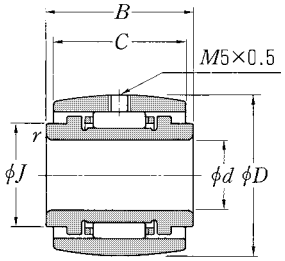
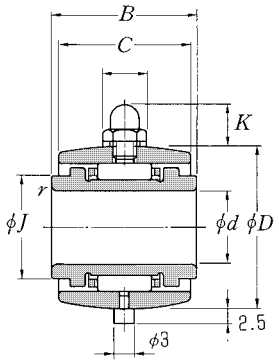



| Bearing numbers | | | Abutment dimensions | | | | Mass |
|---|---|---|---|---|---|---|---|
| bearing | bearing with saddle | | mm | | | | kg |
| | Type SA | Type SB | $L_1$ | $d_1$ | $G$ | $H$ | bearing with saddle (approx.) |
| FRIS16.5 | FRIS16.5SA | FRIS16.5SB | 22 | 21 | 13 | 24～26 | 0.059 |
| FRIS19 | FRIS19SA | FRIS19SB | 22 | 24 | 15 | 27～29 | 0.081 |
| FRIS19-5 | FRIS19-5SA | FRIS19-5SB | 22 | 26 | 15 | 30～32 | 0.120 |
| FRIS22-2 | FRIS22-2SA | — — | 26 | 30 | 15 | 35～37 | 0.208 |
| FRIS25 | FRIS25SA | — — | 26 | 33 | 15 | 37～39 | 0.226 |

## Type FRIS Series B
## (For fine spinning machine/ roving frame)




Type FRIS



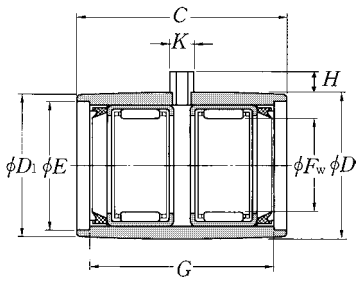

Type FRIS··NP
(With grease nipple/knock pin) [2)]

### d 15～25mm

| Boundary dimensions mm | | | | | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Bearing numbers | | Mass (approx.) kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $d$ $^{0}_{-0.010}$ | $D$ | $B$ $^{0}_{-0.050}$ | $C$ $^{0}_{-0.120}$ | $J$ | $r$s min [1)] | $K$ | | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | Type FRIS | Type FRIS··NP | Type FRIS··NP |
| **15** | 28 $^{0}_{-0.020}$ | 24 | 22 | 22 | 0.3 | 7.7 | | 10 200 | 14 600 | 1 040 | 1 480 | **FRIS15** | **FRIS15NP** | 0.070 |
| **15.875** | 30 $^{0}_{-0.020}$ | 25.4 | 22 | 22 | 0.3 | 7.1 | | 10 200 | 14 600 | 1 040 | 1 480 | **FRIS15.87** | **FRIS15.87NP** | 0.082 |
| **16** | 28 $^{0}_{-0.020}$ | 26 | 22 | 22 | 0.3 | 7.7 | | 10 200 | 14 600 | 1 040 | 1 480 | **FRIS16** | **FRIS16NP** | 0.083 |
| | 30 | 26 | 22 | 22 | 0.3 | 7.1 | | 10 200 | 14 600 | 1 040 | 1 480 | **FRIS16-2** | **FRIS16-2NP** | 0.098 |
| **18** | 31.750 $^{0}_{-0.025}$ | 24 | 23 | 24 | 0.3 | 7.7 | | 11 800 | 18 500 | 1 200 | 1 880 | **FRIS18** | **FRIS18NP** | 0.083 |
| | 32 | 26 | 23 | 24 | 0.3 | 7.7 | | 11 800 | 18 500 | 1 200 | 1 880 | **FRIS18-2** | **FRIS18-2NP** | 0.088 |
| **19.050** | 31.750 $^{0}_{-0.025}$ | 23.81 | 23 | 24 | 0.3 | 7.7 | | 11 800 | 18 500 | 1 200 | 1 880 | **FRIS19.05** | **FRIS19.05NP** | 0.078 |
| | 34 | 25.4 | 23 | 26 | 0.3 | 7.7 | | 12 700 | 17 600 | 1 300 | 1 800 | **FRIS19.05-1** | **FRIS19.05-1NP** | 0.098 |
| **20** | 34 | 24 | 23 | 26 | 0.3 | 7.7 | | 12 700 | 17 600 | 1 300 | 1 800 | **FRIS20-2** | **FRIS20-2NP** | 0.089 |
| | 34 $^{0}_{-0.025}$ | 26 | 23 | 26 | 0.3 | 7.7 | | 12 700 | 17 600 | 1 300 | 1 800 | **FRIS20-4** | **FRIS20-4NP** | 0.097 |
| | 36 | 26 | 23 | 26 | 0.3 | 7.1 | | 12 700 | 17 600 | 1 300 | 1 800 | **FRIS20-7** | **FRIS20-7NP** | 0.125 |
| **22** | 40 $^{0}_{-0.025}$ | 26 | 24 | 30 | 0.6 | 7.1 | | 16 200 | 22 300 | 1 650 | 2 280 | **FRIS22-1** | **FRIS22-1NP** | 0.154 |
| **22.225** | 40 $^{0}_{-0.025}$ | 25.4 | 24 | 30 | 0.6 | 7.1 | | 16 200 | 22 300 | 1 650 | 2 280 | **FRIS22.22** | **FRIS22.22NP** | 0.145 |
| **25** | 42 $^{0}_{-0.025}$ | 26 | 24 | 32 | 0.6 | 7.1 | | 16 000 | 22 500 | 1 640 | 2 290 | **FRIS25-1** | **FRIS25-1NP** | 0.154 |
| | 44 | 26 | 24 | 34 | 0.6 | 7.1 | | 17 200 | 25 100 | 1 750 | 2 560 | **FRIS25-2** | **FRIS25-2NP** | 0.174 |

Note 1) Allowable minimum chamfer dimension $r$.
2) Example of bearing with grease nipple: Ex. FRIS 18N
Bearing with knock pin only: Ex. FRIS 15.87P

## Type FR
## (Drawing frame)




Type FR

$F_w$ 17.462～22mm

| Boundary dimensions mm | | | | | | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Bearing numbers | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | | $D$ $^{0}_{-0.050}$ | $C$ $^{0}_{-0.20}$ | $E$ | $G$ | $H$ | $K$ | $D_1$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | (approx.) |
| | | 26.5 | 46 | 24 | 31 | 5 | 5 | 26 | 10 200 | 16 600 | 1 040 | 1 690 | **FR17.46P** | 0.066 |
| **17.462** | $^{+0.053}_{+0.010}$ | 26.5 | 47.6 | 24.6 | 42.86 | 4.76 | 4.76 | 26 | 14 400 | 25 900 | 1 470 | 2 640 | **FR17.46-1P** | 0.075 |
| | | 26.5 | 52.38 | 24.6 | 42.86 | 4.76 | 4.76 | 26 | 14 400 | 25 900 | 1 470 | 2 640 | **FR17.46-2P** | 0.082 |
| | | 26.988 | 41.28 | 24 | 31 | 5 | 5 | 26.5 | 10 200 | 16 600 | 1 040 | 1 690 | **FR17.46-3P** | 0.068 |
| **19.050** | $^{+0.053}_{+0.020}$ | 31.750 | 52.38 | 28 | 42.86 | 4.76 | 4.76 | 31.2 | 16 700 | 25 800 | 1 700 | 2 630 | **FR19.05P** | 0.134 |
| **22** | $^{+0.053}_{+0.020}$ | 34 | 46 | 28 | 43 | 4.76 | 4.76 | 33.5 | 23 400 | 44 000 | 2 380 | 4 500 | **FR22P** | 0.150 |

# Tensioner Pulleys for Textile Machinery

These pulleys are used to guide and tension the tapes and belts driving the spindles of a fine spinning machine, a roving frame, a false twister, etc.

A pulley drawn precisely from steel plate by precision deep drawing is press-fitted in the outer ring of shaft bearing in in place of inner ring.

## Types and construction

Single-row or double-row ball bearing is built in this pulley, which is internally prefilled with lithium soap base grease. Grease is replenished through a grease hole which is internally provided in the stud. Two different mounting methods are available as follows for these pulleys; one method is to bolt a pulley to the roller carrier of machine with holder bolt using a special-purposed holder (Type code: **JF**··**S**) and another method is to bolt directly a pulley to machine frame using the bolting hole drilled in the stud and a corresponding hexagon head bolt. A knock pin press-fitted in the stud end face is to lock the bolt after tightened.

When a pulley is directly bolted to machine frame, provide the hexagon head bolt center with a grease through-hole for grease replenishing and screw a grease nipple in its end face.

In mounting a double-row ball bearing type tension pulley, insert the stud directly into machine frame and tighten it with nut, without using the special-purposed holder. Replenish grease using the box nut.

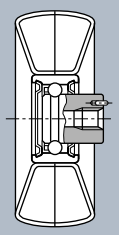


**Fig. 1**
**Single-row bearing type ension pulley**

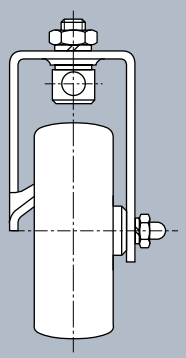


**Fig. 2**
**Tension pulley with holder**

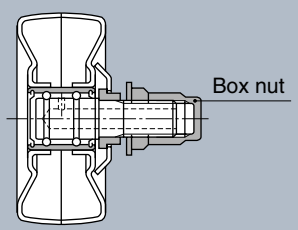



**Fig. 3**
**Double-row bearing type tension pulley**

## Composition of nominal number

Nominal number comprises type code (**JPU**··**S**) , dimension code (Pulley outer dia. ($D$) and tail code. Tension pulley with special-purposed holder is represented by nominal number + tail code (holder nominal number).

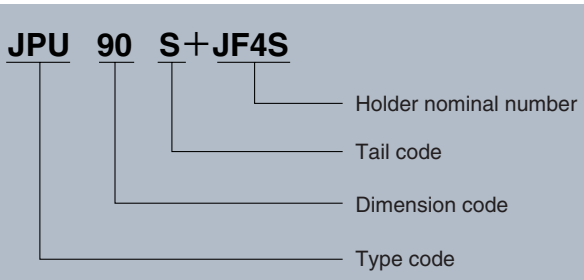
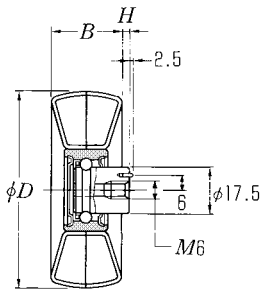
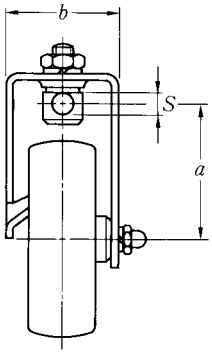


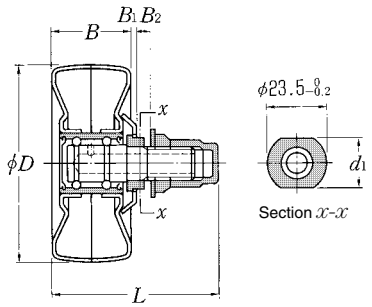

**Fig. 4**




Example 1




Example 2




Example 3

## *D* 60～100mm

| Boundary dimensions mm | | | | | | | | | | Bearing numbers | Sample number | Mass kg (approx.) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *D* | *B* | $B_1$ | *H* | $d_1$ $^{0}_{-0.3}$ | *a* | *b* | *S* | *L* | $B_2$ | | | |
| **60** | 25 | — | 3.5 | — | — | — | — | — | — | **JPU60S** | 1 | 0.140 |
| **70** | 25 | — | 3.5 | — | — | — | — | — | — | **JPU70S** | 1 | 0.160 |
| | 25 | — | 3.5 | — | 48 | 39 | 10.2 | — | — | **JPU70S＋JF1S** | 2 | 0.235 |
| | 30 | 2 | — | 21 | — | — | — | 67 ～71 | 5～9 | **JPU70-2S** | 3 | 0.245 |
| | 32 | — | — | — | — | — | — | — | — | **JPU70-1S** | 1 | 0.175 |
| | 32 | — | — | — | 48 | 46 | 10.2 | — | — | **JPU70-1S＋JF2S** | 2 | 0.250 |
| **90** | 25 | — | 3.5 | — | — | — | — | — | — | **JPU90S** | 1 | 0.215 |
| | 25 | — | 3.5 | — | 58 | 39 | 10.2 | — | — | **JPU90S＋JF3S** | 2 | 0.300 |
| | 32 | — | — | — | — | — | — | — | — | **JPU90-1S** | 1 | 0.230 |
| | 32 | — | — | — | 58 | 46 | 10.2 | — | — | **JPU90-1S＋JF4S** | 2 | 0.320 |
| | 32 | 3.5 | — | 21 | — | — | — | 70.5～74.5 | 5～9 | **JPU90-9S** | 3 | 0.325 |
| | 45 | 3.5 | — | 21 | — | — | — | 83.5～87.5 | 5～9 | **JPU90-10S** | 3 | 0.380 |
| **100** | 32 | 3.5 | — | 21 | — | — | — | 70.5～74.5 | 5～9 | **JPU100-14S** | 3 | 0.290 |
| | 40 | 3.5 | — | 21 | — | — | — | 78.5～82.5 | 5～9 | **JPU100-12S** | 3 | 0.390 |

## Contents of Appendix


Approximate conversion values against Rockwell C hardness of steel materials ···C-2
˚C-˚F temperature conversion table ·····C-3
Approximate conversion values against Vickers hardness of steel materials ···C-4
Contrast table of SI and CGS system units, gravitation system units ·····C-6
Conversion to SI units ·····C-6
Integer multiplication of SI unit 10 ·····C-7
kgf-N conversion table·····C-8
Viscosity conversion table ·····C-9
Ordinary tolerance for cutting dimensions ·····C-10
Ordinary tolerances for castings ·····C-10
Ordinary tolerances for press-formed metal parts ·····C-10
Ordinary tolerance for die casts ·····C-11
Ordinary tolerances for sheared metal plates ·····C-12
Ordinary tolerance for sintered metal parts ·····C-13
Ordinary tolerance for steel castings ·····C-14
Ordinary tolerance for aluminum alloy castings ·····C-14
Ordinary tolerance for length ·····C-15
Ordinary tolerance for wall thickness ·····C-15
Maximum draft angle value ·····C-15
Tolerance for shaft diameter ·····C-16
Dimensional tolerance for housing bore·····C-18
Inch - Millimeter conversion table ·····C-20

Approximate conversion values against Rockwell C hardness of steel materials[1)]

| Rockwell C scale hardness[2)] | Pickers hardness (DPH) | Brinnel hardness 100 mm ball, load 3000 kgf | | | Rockwell hardness[2)] | | | Rockwell special hardness special brake indenter | | | Shore hardness | Tensile strength kgf/mm² Approx. value | Rockwell C scale hardness[2)] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Standard ball | Hultgren ball | Tungsten carbite ball | A scale Load 60kgf brale identer | B scale Load 100 kgf dia. 1/16in. ball | D-scale Load 100 kgf brale indenter | 15-N scale Load 15 kgfkgf | 30-N scale Load 30 kgf | 45-N scale load 45 kgf | | | |
| 68 | 940 | — | — | — | 85.6 | — | 76.9 | 93.2 | 84.4 | 75.4 | 97 | — | 68 |
| 67 | 900 | — | — | — | 85.0 | — | 76.1 | 92.9 | 83.6 | 74.2 | 95 | — | 67 |
| 66 | 865 | — | — | — | 84.5 | — | 75.4 | 92.5 | 82.8 | 73.3 | 92 | — | 66 |
| 65 | 832 | — | — | 739 | 83.9 | — | 74.5 | 92.2 | 81.9 | 72.0 | 91 | — | 65 |
| 64 | 800 | — | — | 722 | 83.4 | — | 73.8 | 91.8 | 81.1 | 71.0 | 88 | — | 64 |
| 63 | 772 | — | — | 705 | 82.8 | — | 73.0 | 91.4 | 80.1 | 69.9 | 87 | — | 63 |
| 62 | 746 | — | — | 688 | 82.3 | — | 72.2 | 91.1 | 79.3 | 68.8 | 85 | — | 62 |
| 61 | 720 | — | — | 670 | 81.8 | — | 71.5 | 90.7 | 78.4 | 67.7 | 83 | — | 61 |
| 60 | 697 | — | 613 | 654 | 81.2 | — | 70.7 | 90.2 | 77.5 | 66.6 | 81 | — | 60 |
| 59 | 674 | — | 599 | 634 | 80.7 | — | 69.9 | 89.8 | 76.6 | 65.5 | 80 | — | 59 |
| 58 | 653 | — | 587 | 615 | 80.1 | — | 69.2 | 89.3 | 75.7 | 64.3 | 78 | — | 58 |
| 57 | 633 | — | 575 | 595 | 79.6 | — | 68.5 | 88.9 | 74.8 | 63.2 | 76 | — | 57 |
| 56 | 613 | — | 561 | 577 | 79.0 | — | 67.7 | 88.3 | 73.9 | 62.0 | 75 | — | 56 |
| 55 | 595 | — | 546 | 560 | 78.5 | — | 66.9 | 87.9 | 73.0 | 60.9 | 74 | 212 | 55 |
| 54 | 577 | — | 534 | 543 | 78.0 | — | 66.1 | 87.4 | 72.0 | 59.8 | 72 | 205 | 54 |
| 53 | 560 | — | 519 | 525 | 77.4 | — | 65.4 | 86.9 | 71.2 | 58.6 | 71 | 199 | 53 |
| 52 | 544 | 500 | 508 | 512 | 76.8 | — | 64.6 | 86.4 | 70.2 | 57.4 | 69 | 192 | 52 |
| 51 | 528 | 487 | 494 | 496 | 76.3 | — | 63.8 | 85.9 | 69.4 | 56.1 | 68 | 186 | 51 |
| 50 | 513 | 475 | 481 | 481 | 75.9 | — | 63.1 | 85.5 | 68.5 | 55.0 | 67 | 179 | 50 |
| 49 | 498 | 464 | 469 | 469 | 75.2 | — | 62.1 | 85.0 | 67.6 | 53.8 | 66 | 172 | 49 |
| 48 | 484 | 451 | 455 | 455 | 74.7 | — | 61.4 | 84.5 | 66.7 | 52.5 | 64 | 167 | 48 |
| 47 | 471 | 442 | 443 | 443 | 74.1 | — | 60.8 | 83.9 | 65.8 | 51.4 | 63 | 161 | 47 |
| 46 | 458 | 432 | 432 | 432 | 73.6 | — | 60.0 | 83.5 | 64.8 | 50.3 | 62 | 156 | 46 |
| 45 | 446 | 421 | 421 | 421 | 73.1 | — | 59.2 | 83.0 | 64.0 | 49.0 | 60 | 151 | 45 |
| 44 | 434 | 409 | 409 | 409 | 72.5 | — | 58.5 | 82.5 | 63.1 | 47.8 | 58 | 146 | 44 |
| 43 | 423 | 400 | 400 | 400 | 72.0 | — | 57.7 | 82.0 | 62.2 | 46.7 | 57 | 141 | 43 |
| 42 | 412 | 390 | 390 | 390 | 71.5 | — | 56.9 | 81.5 | 61.3 | 45.5 | 56 | 136 | 42 |
| 41 | 402 | 381 | 381 | 381 | 70.9 | — | 56.2 | 80.9 | 60.4 | 44.3 | 55 | 132 | 41 |
| 40 | 392 | 371 | 371 | 371 | 70.4 | — | 55.4 | 80.4 | 59.5 | 43.1 | 54 | 127 | 40 |
| 39 | 382 | 362 | 362 | 362 | 69.9 | — | 54.6 | 79.9 | 58.6 | 41.9 | 52 | 124 | 39 |
| 38 | 372 | 353 | 353 | 353 | 69.4 | — | 53.8 | 79.4 | 57.7 | 40.8 | 51 | 120 | 38 |
| 37 | 363 | 344 | 344 | 344 | 68.9 | — | 53.1 | 78.8 | 56.8 | 39.6 | 50 | 118 | 37 |
| 36 | 354 | 336 | 336 | 336 | 68.4 | (109.0) | 52.3 | 78.3 | 55.9 | 38.4 | 49 | 114 | 36 |
| 35 | 345 | 327 | 327 | 327 | 67.9 | (108.5) | 51.5 | 77.7 | 55.0 | 37.2 | 48 | 110 | 35 |
| 34 | 336 | 319 | 319 | 319 | 67.4 | (108.0) | 50.8 | 77.2 | 54.2 | 36.1 | 47 | 108 | 34 |
| 33 | 327 | 311 | 311 | 311 | 66.8 | (107.5) | 50.0 | 76.6 | 53.3 | 34.9 | 46 | 105 | 33 |
| 32 | 318 | 301 | 301 | 301 | 66.3 | (107.0) | 49.2 | 76.1 | 52.1 | 33.7 | 44 | 102 | 32 |
| 31 | 310 | 294 | 294 | 294 | 65.8 | (106.0) | 48.4 | 75.6 | 51.3 | 32.5 | 43 | 100 | 31 |
| 30 | 302 | 286 | 286 | 286 | 65.3 | (105.5) | 47.7 | 75.0 | 50.4 | 31.3 | 42 | 97 | 30 |
| 29 | 294 | 279 | 279 | 279 | 64.7 | (104.5) | 47.0 | 74.5 | 49.5 | 30.1 | 41 | 95 | 29 |
| 28 | 286 | 271 | 271 | 271 | 64.3 | (104.0) | 46.1 | 73.9 | 48.6 | 28.9 | 41 | 93 | 28 |
| 27 | 279 | 264 | 264 | 264 | 63.8 | (103.0) | 45.2 | 73.3 | 47.7 | 27.8 | 40 | 90 | 27 |
| 26 | 272 | 258 | 258 | 258 | 63.3 | (102.5) | 44.6 | 72.8 | 46.8 | 26.7 | 38 | 88 | 26 |
| 25 | 266 | 253 | 253 | 253 | 62.8 | (101.5) | 43.8 | 72.2 | 45.9 | 25.5 | 38 | 86 | 25 |
| 24 | 260 | 247 | 247 | 247 | 62.4 | (101.0) | 43.1 | 71.6 | 45.0 | 24.3 | 37 | 84 | 24 |
| 23 | 254 | 243 | 243 | 243 | 62.0 | 100.0 | 42.1 | 71.0 | 44.0 | 23.1 | 36 | 82 | 23 |
| 22 | 248 | 237 | 237 | 237 | 61.5 | 99.0 | 41.6 | 70.5 | 43.2 | 22.0 | 35 | 80 | 22 |
| 21 | 243 | 231 | 231 | 231 | 61.0 | 98.5 | 40.9 | 69.9 | 42.3 | 20.7 | 35 | 79 | 21 |
| 20 | 238 | 226 | 226 | 226 | 60.5 | 97.8 | 40.1 | 69.4 | 41.5 | 19.6 | 34 | 77 | 20 |
| (18) | 230 | 219 | 219 | 219 | — | 96.7 | — | — | — | — | 33 | 75 | (18) |
| (16) | 222 | 212 | 212 | 212 | — | 95.5 | — | — | — | — | 32 | 72 | (16) |
| (14) | 213 | 203 | 203 | 203 | — | 93.9 | — | — | — | — | 31 | 69 | (14) |
| (12) | 204 | 194 | 194 | 194 | — | 92.3 | — | — | — | — | 29 | 66 | (12) |
| (10) | 196 | 187 | 187 | 187 | — | 90.7 | — | — | — | — | 28 | 63 | (10) |
| ( 8) | 188 | 179 | 179 | 179 | — | 89.5 | — | — | — | — | 27 | 61 | ( 8) |
| ( 6) | 180 | 171 | 171 | 171 | — | 87.1 | — | — | — | — | 26 | 59 | ( 6) |
| ( 4) | 173 | 165 | 165 | 165 | — | 85.5 | — | — | — | — | 25 | 56 | ( 4) |
| ( 2) | 166 | 158 | 158 | 158 | — | 83.5 | — | — | — | — | 24 | 54 | ( 2) |
| ( 0) | 160 | 152 | 152 | 152 | — | 81.7 | — | — | — | — | 24 | 53 | ( 0) |

Note 1) Approximate values were determined from “Conversion Tables” of **JIS Z 8413** and **Z 8438**.
2) Some of the parenthesized values in the above table are not used so frequently.

˚C-˚F temperature conversion table

| ˚C | | F | ˚C | | F | ˚C | | F | ˚C | | F |
|---|---|---|---|---|---|---|---|---|---|---|---|
| −73 | −100 | −148 | −1.6 | 29 | 84.2 | 17.7 | 64 | 147.2 | 37.1 | 99 | 210.2 |
| −62 | − 80 | −112 | −1.1 | 30 | 86.0 | 18.2 | 65 | 149.0 | 37.7 | 100 | 212 |
| −51 | − 60 | − 76 | −0.6 | 31 | 87.8 | 18.8 | 66 | 150.8 | 40.6 | 105 | 221 |
| −40 | − 40 | − 40 | 0 | 32 | 89.6 | 19.3 | 67 | 152.6 | 43 | 110 | 230 |
| −29 | − 20 | − 4 | 0.5 | 33 | 91.4 | 19.9 | 68 | 154.4 | 49 | 120 | 248 |
| −23.3 | − 10 | 14 | 1.1 | 34 | 93.2 | 20.4 | 69 | 156.2 | 54 | 130 | 266 |
| −17.7 | 0 | 32 | 1.6 | 35 | 95.0 | 21.0 | 70 | 158.0 | 60 | 140 | 284 |
| −17.2 | 1 | 33.8 | 2.2 | 36 | 96.8 | 21.5 | 71 | 159.8 | 65 | 150 | 302 |
| −16.6 | 2 | 35.6 | 2.7 | 37 | 98.6 | 22.2 | 72 | 161.6 | 71 | 160 | 320 |
| −16.1 | 3 | 37.4 | 3.3 | 38 | 100.4 | 22.7 | 73 | 163.4 | 76 | 170 | 338 |
| −15.5 | 4 | 39.2 | 3.8 | 39 | 102.2 | 23.3 | 74 | 165.2 | 83 | 180 | 356 |
| −15.0 | 5 | 41.0 | 4.4 | 40 | 104.0 | 23.8 | 75 | 167.0 | 88 | 190 | 374 |
| −14.4 | 6 | 42.8 | 4.9 | 41 | 105.8 | 24.4 | 76 | 168.8 | 93 | 200 | 392 |
| −13.9 | 7 | 44.6 | 5.4 | 42 | 107.6 | 25.0 | 77 | 170.6 | 121 | 250 | 482 |
| −13.3 | 8 | 46.4 | 6.0 | 43 | 109.4 | 25.5 | 78 | 172.4 | 149 | 300 | 572 |
| −12.7 | 9 | 48.2 | 6.6 | 44 | 111.2 | 26.2 | 79 | 174.2 | 177 | 350 | 662 |
| −12.2 | 10 | 50.0 | 7.1 | 45 | 113.0 | 26.8 | 80 | 176.0 | 204 | 400 | 752 |
| −11.6 | 11 | 51.8 | 7.7 | 46 | 114.8 | 27.3 | 81 | 177.8 | 232 | 450 | 842 |
| −11.1 | 12 | 53.6 | 8.2 | 47 | 116.6 | 27.7 | 82 | 179.6 | 260 | 500 | 932 |
| −10.5 | 13 | 55.4 | 8.8 | 48 | 118.4 | 28.2 | 83 | 181.4 | 288 | 550 | 1 022 |
| −10.0 | 14 | 57.2 | 9.3 | 49 | 120.2 | 28.8 | 84 | 183.2 | 315 | 600 | 1 112 |
| − 9.4 | 15 | 59.0 | 9.9 | 50 | 122.0 | 29.3 | 85 | 185.0 | 343 | 650 | 1 202 |
| − 8.8 | 16 | 61.8 | 10.4 | 51 | 123.8 | 29.9 | 86 | 186.8 | 371 | 700 | 1 292 |
| − 8.3 | 17 | 63.6 | 11.1 | 52 | 125.6 | 30.4 | 87 | 188.6 | 399 | 750 | 1 382 |
| − 7.7 | 18 | 65.4 | 11.5 | 53 | 127.4 | 31.0 | 88 | 190.4 | 426 | 800 | 1 472 |
| − 7.2 | 19 | 67.2 | 12.1 | 54 | 129.2 | 31.5 | 89 | 192.2 | 454 | 850 | 1 562 |
| − 6.6 | 20 | 68.0 | 12.6 | 55 | 131.0 | 32.1 | 90 | 194.0 | 482 | 900 | 1 652 |
| − 6.1 | 21 | 69.8 | 13.2 | 56 | 132.8 | 32.6 | 91 | 195.8 | 510 | 950 | 1 742 |
| − 5.5 | 22 | 71.6 | 13.7 | 57 | 134.6 | 33.3 | 92 | 197.6 | 538 | 1 000 | 1 832 |
| − 5.0 | 23 | 73.4 | 14.3 | 58 | 136.4 | 33.8 | 93 | 199.4 | 593 | 1 100 | 2 012 |
| − 4.4 | 24 | 75.2 | 14.8 | 59 | 138.2 | 34.4 | 94 | 201.2 | 648 | 1 200 | 2 192 |
| − 3.9 | 25 | 77.0 | 15.6 | 60 | 140.0 | 34.9 | 95 | 203.0 | 704 | 1 300 | 2 372 |
| − 3.3 | 26 | 78.8 | 16.1 | 61 | 141.8 | 35.5 | 96 | 204.8 | 760 | 1 400 | 2 552 |
| − 2.8 | 27 | 80.6 | 16.6 | 62 | 143.6 | 36.1 | 97 | 206.6 | 815 | 1 500 | 2 732 |
| − 2.2 | 28 | 82.4 | 17.1 | 63 | 145.4 | 36.6 | 98 | 208.4 | 871 | 1 600 | 2 937 |

(How to read this table)
For example, for conversion from 38˚C to ˚F read 38 in 4th row box (10th line from the top) and then read the right-side ˚F box.
38˚C is equivalent to 100.4˚F upon reading. Furthermore, for conversion from 38˚F to ˚C read the left-side ˚C box.
˚F38 is equivalent to 3.3˚C upon reading.

$C = 5/9(F - 32)$
$F = 9/5C + 32$

Approximate conversion values against Vickers hardness of steel materials

| Vickers hardness (DPH) | Brinnel hardness 10 mm ball, load 3000 kgf | | | Rockwell hardness[2)] | | | | Rockwell special hardness special brake indenter | | | Shore hardness | Tensile strength kgf/mm² Approx. value | Vickers hardness Load 50kgf |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Standard ball | Hultgren ball | Tungsten carbite ball | A scale Load 60 kgf brale identer | B scale Load 100 kgf dia. 1/16in. ball | C-scale Load 150 kgf brale indenter | D-scale Load 100 kgf brale indenter | 15-N scale Load 15 kgf | 30-N scale Load 30 kgf | 45-N scale load 45 kgf | | | |
| 940 | — | — | — | 85.6 | — | 68.0 | 76.9 | 93.2 | 84.4 | 75.4 | 97 | — | 940 |
| 920 | — | — | — | 85.3 | — | 67.5 | 76.5 | 93.0 | 84.0 | 74.8 | 96 | — | 920 |
| 900 | — | — | — | 85.0 | — | 67.0 | 76.1 | 92.9 | 83.6 | 74.2 | 95 | — | 900 |
| 880 | — | — | 767 | 84.7 | — | 66.4 | 75.7 | 92.7 | 83.1 | 73.6 | 93 | — | 880 |
| 860 | — | — | 757 | 84.4 | — | 65.9 | 75.3 | 92.5 | 82.7 | 73.1 | 92 | — | 860 |
| 840 | — | — | 745 | 84.1 | — | 65.3 | 74.8 | 92.3 | 82.2 | 72.2 | 91 | — | 840 |
| 820 | — | — | 733 | 83.8 | — | 64.7 | 74.3 | 92.1 | 81.7 | 71.8 | 90 | — | 820 |
| 800 | — | — | 722 | 83.4 | — | 64.0 | 73.8 | 91.8 | 81.1 | 71.0 | 88 | — | 800 |
| 780 | — | — | 710 | 83.0 | — | 63.3 | 73.3 | 91.5 | 80.4 | 70.2 | 87 | — | 780 |
| 760 | — | — | 698 | 82.6 | — | 62.5 | 72.6 | 91.2 | 79.7 | 69.4 | 86 | — | 760 |
| 740 | — | — | 684 | 82.2 | — | 61.8 | 72.1 | 91.0 | 79.1 | 68.6 | 84 | — | 740 |
| 720 | — | — | 670 | 81.8 | — | 61.0 | 71.5 | 90.7 | 78.4 | 67.7 | 83 | — | 720 |
| 700 | — | 615 | 656 | 81.3 | — | 60.1 | 70.8 | 90.3 | 77.6 | 66.7 | 81 | — | 700 |
| 690 | — | 610 | 647 | 81.1 | — | 59.7 | 70.5 | 90.1 | 77.2 | 66.2 | — | — | 690 |
| 680 | — | 603 | 638 | 80.8 | — | 59.2 | 70.1 | 89.8 | 76.8 | 65.7 | 80 | — | 680 |
| 670 | — | 597 | 630 | 80.5 | — | 58.8 | 69.8 | 89.7 | 76.4 | 65.3 | — | — | 670 |
| 660 | — | 590 | 620 | 80.3 | — | 58.3 | 69.4 | 89.5 | 75.9 | 64.7 | 79 | — | 660 |
| 650 | — | 585 | 611 | 80.0 | — | 57.8 | 69.0 | 89.2 | 75.5 | 64.1 | — | — | 650 |
| 640 | — | 578 | 601 | 79.8 | — | 57.3 | 68.7 | 89.0 | 75.1 | 63.5 | 77 | — | 640 |
| 630 | — | 571 | 591 | 79.5 | — | 56.8 | 68.3 | 88.8 | 74.6 | 63.0 | — | — | 630 |
| 620 | — | 564 | 582 | 79.2 | — | 56.3 | 67.9 | 88.5 | 74.2 | 62.4 | 75 | — | 620 |
| 610 | — | 557 | 573 | 78.9 | — | 55.7 | 67.5 | 88.2 | 73.6 | 61.7 | — | — | 610 |
| 600 | — | 550 | 564 | 78.6 | — | 55.2 | 67.0 | 88.0 | 73.2 | 61.2 | 74 | — | 600 |
| 590 | — | 542 | 554 | 78.4 | — | 54.7 | 66.7 | 87.8 | 72.7 | 60.5 | — | 210 | 590 |
| 580 | — | 535 | 545 | 78.0 | — | 54.1 | 66.2 | 87.5 | 72.1 | 59.9 | 72 | 206 | 580 |
| 570 | — | 527 | 535 | 77.8 | — | 53.6 | 65.8 | 87.2 | 71.7 | 59.3 | — | 202 | 570 |
| 560 | — | 519 | 525 | 77.4 | — | 53.0 | 65.4 | 86.9 | 71.2 | 58.6 | 71 | 199 | 560 |
| 550 | 505 | 512 | 517 | 77.0 | — | 52.3 | 64.8 | 86.6 | 70.5 | 57.8 | — | 195 | 550 |
| 540 | 496 | 503 | 507 | 76.7 | — | 51.7 | 64.4 | 86.3 | 70.0 | 57.0 | 69 | 190 | 540 |
| 530 | 488 | 495 | 497 | 76.4 | — | 51.1 | 63.9 | 86.0 | 69.5 | 56.2 | — | 186 | 530 |
| 520 | 480 | 487 | 488 | 76.1 | — | 50.5 | 63.5 | 85.7 | 69.0 | 55.6 | 67 | 183 | 520 |
| 510 | 473 | 479 | 479 | 75.7 | — | 49.8 | 62.9 | 85.4 | 68.3 | 54.7 | — | 179 | 510 |
| 500 | 465 | 471 | 471 | 75.3 | — | 49.1 | 62.2 | 85.0 | 67.7 | 53.9 | 66 | 174 | 500 |
| 490 | 456 | 460 | 460 | 74.9 | — | 48.4 | 61.6 | 84.7 | 67.1 | 53.1 | — | 169 | 490 |
| 480 | 448 | 452 | 452 | 74.5 | — | 47.7 | 61.3 | 84.3 | 66.4 | 52.2 | 64 | 165 | 480 |
| 470 | 441 | 442 | 442 | 74.1 | — | 46.9 | 60.7 | 83.9 | 65.7 | 51.3 | — | 160 | 470 |
| 460 | 433 | 433 | 433 | 73.6 | — | 46.1 | 60.1 | 83.6 | 64.9 | 50.4 | 62 | 156 | 460 |
| 450 | 425 | 425 | 425 | 73.3 | — | 45.3 | 59.4 | 83.2 | 64.3 | 49.4 | — | 153 | 450 |
| 440 | 415 | 415 | 415 | 72.8 | — | 44.5 | 58.8 | 82.8 | 63.5 | 48.4 | 59 | 149 | 440 |
| 430 | 405 | 405 | 405 | 72.3 | — | 43.6 | 58.2 | 82.3 | 62.7 | 47.4 | — | 144 | 430 |
| 420 | 397 | 397 | 397 | 71.8 | — | 42.7 | 57.5 | 81.8 | 61.9 | 46.4 | 57 | 140 | 420 |
| 410 | 388 | 388 | 388 | 71.4 | — | 41.8 | 56.8 | 81.4 | 61.1 | 45.3 | — | 136 | 410 |
| 400 | 379 | 379 | 379 | 70.8 | — | 40.8 | 56.0 | 81.0 | 60.2 | 44.1 | 55 | 131 | 400 |
| 390 | 369 | 369 | 369 | 70.3 | — | 39.8 | 55.2 | 80.3 | 59.3 | 42.9 | — | 127 | 390 |
| 380 | 360 | 360 | 360 | 69.8 | (110.0) | 38.8 | 54.4 | 79.8 | 58.4 | 41.7 | 52 | 123 | 380 |

continued

| Vickers hardness (DPH) | Brinnel hardness 10 mm ball, load 3000 kgf | | | Rockwell hardness[2] | | | | Rockwell special hardness special brake indenter | | | Shore hardness | Tensile strength kgf/mm² Approx. value | Vickers hardness Load 50kgf |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Standard ball | Hultgren ball | Tungsten carbite ball | A scale Load 60 kgf brale identer | B scale Load 100 kgf dia. 1/16in. ball | C-scale Load 150 kgf brale indenter | D-scale Load 100 kgf brale indenter | 15-N scale Load 15 kgf | 30-N scale Load 30 kgf | 45-N scale load 45 kgf | | | |
| 370 | 350 | 350 | 350 | 69.2 | — | 37.7 | 53.6 | 79.2 | 57.4 | 40.4 | — | 120 | 370 |
| 360 | 341 | 341 | 341 | 68.7 | (109.0) | 36.6 | 52.8 | 78.6 | 56.4 | 39.1 | 50 | 115 | 360 |
| 350 | 331 | 331 | 331 | 68.1 | — | 35.5 | 51.9 | 78.0 | 55.4 | 37.8 | — | 112 | 350 |
| 340 | 322 | 322 | 322 | 67.6 | (108.0) | 34.4 | 51.1 | 77.4 | 54.4 | 36.5 | 47 | 109 | 340 |
| 330 | 313 | 313 | 313 | 67.0 | — | 33.3 | 50.2 | 76.8 | 53.6 | 35.2 | — | 105 | 330 |
| 320 | 303 | 303 | 303 | 66.4 | (107.0) | 32.2 | 49.4 | 76.2 | 52.3 | 33.9 | 45 | 103 | 320 |
| 310 | 294 | 294 | 294 | 65.8 | — | 31.0 | 48.4 | 75.6 | 51.3 | 32.5 | — | 100 | 310 |
| 300 | 284 | 284 | 284 | 65.2 | (105.5) | 29.8 | 47.5 | 74.9 | 50.2 | 31.1 | 42 | 97 | 300 |
| 295 | 280 | 280 | 280 | 64.8 | — | 29.2 | 47.1 | 74.6 | 49.7 | 30.4 | — | 96 | 295 |
| 290 | 275 | 275 | 275 | 64.5 | (104.5) | 28.5 | 46.5 | 74.2 | 49.0 | 29.5 | 41 | 94 | 290 |
| 285 | 270 | 270 | 270 | 64.2 | — | 27.8 | 46.0 | 73.8 | 48.4 | 28.7 | — | 92 | 285 |
| 280 | 265 | 265 | 265 | 63.8 | (103.5) | 27.1 | 45.3 | 73.4 | 47.8 | 27.9 | 40 | 91 | 280 |
| 275 | 261 | 261 | 261 | 63.5 | — | 26.4 | 44.9 | 73.0 | 47.2 | 27.1 | — | 89 | 275 |
| 270 | 256 | 256 | 256 | 63.1 | (102.0) | 25.6 | 44.3 | 72.6 | 46.4 | 26.2 | 38 | 87 | 270 |
| 265 | 252 | 252 | 252 | 62.7 | — | 24.8 | 43.7 | 72.1 | 45.7 | 25.2 | — | 86 | 265 |
| 260 | 247 | 247 | 247 | 62.4 | (101.0) | 24.0 | 43.1 | 71.6 | 45.0 | 24.3 | 37 | 84 | 260 |
| 255 | 243 | 243 | 243 | 62.0 | — | 23.1 | 42.2 | 71.1 | 44.2 | 23.2 | — | 82 | 255 |
| 250 | 238 | 238 | 238 | 61.6 | 99.5 | 22.2 | 41.7 | 70.6 | 43.4 | 22.2 | 36 | 81 | 250 |
| 245 | 233 | 233 | 233 | 61.2 | — | 21.3 | 41.1 | 70.1 | 42.5 | 21.1 | — | 79 | 245 |
| 240 | 228 | 228 | 228 | 60.7 | 98.1 | 20.3 | 40.3 | 69.6 | 41.7 | 19.9 | 34 | 78 | 240 |
| 230 | 219 | 219 | 219 | — | 96.7 | (18.0) | — | — | — | — | 33 | 75 | 230 |
| 220 | 209 | 209 | 209 | — | 95.0 | (15.7) | — | — | — | — | 32 | 71 | 220 |
| 210 | 200 | 200 | 200 | — | 93.4 | (13.4) | — | — | — | — | 30 | 68 | 210 |
| 200 | 190 | 190 | 190 | — | 91.5 | (11.0) | — | — | — | — | 29 | 65 | 200 |
| 190 | 181 | 181 | 181 | — | 89.5 | (8.5) | — | — | — | — | 28 | 62 | 190 |
| 180 | 171 | 171 | 171 | — | 87.1 | (6.0) | — | — | — | — | 26 | 59 | 180 |
| 170 | 162 | 162 | 162 | — | 85.0 | (3.0) | — | — | — | — | 25 | 56 | 170 |
| 160 | 152 | 152 | 152 | — | 81.7 | (0.0) | — | — | — | — | 24 | 53 | 160 |
| 150 | 143 | 143 | 143 | — | 78.7 | — | — | — | — | — | 22 | 50 | 150 |
| 140 | 133 | 133 | 133 | — | 75.0 | — | — | — | — | — | 21 | 46 | 140 |
| 130 | 124 | 124 | 124 | — | 71.2 | — | — | — | — | — | 20 | 44 | 130 |
| 120 | 114 | 114 | 114 | — | 66.7 | — | — | — | — | — | — | 40 | 120 |
| 110 | 105 | 105 | 105 | — | 62.3 | — | — | — | — | — | — | — | 110 |
| 100 | 95 | 95 | 95 | — | 56.2 | — | — | — | — | — | — | — | 100 |
| 95 | 90 | 90 | 90 | — | 52.0 | — | — | — | — | — | — | — | 95 |
| 90 | 86 | 86 | 86 | — | 48.0 | — | — | — | — | — | — | — | 90 |
| 85 | 81 | 81 | 81 | — | 41.0 | — | — | — | — | — | — | — | 85 |

1) Some of the parenthesized values in the above table are not used so frequently.
2) Approximate values were determined from “Conversion Tables” of JIS Z 8413 and Z 8438.

Contrast table of SI and CGS system units, gravitation system units

| Unit system \ Quantity | Length $L$ | Mass $M$ | Time $T$ | Acceleration | Force | Stress | Pressure | Energy |
|---|---|---|---|---|---|---|---|---|
| SI | m | kg | s | $m/s^2$ | N | Pa | Pa | J |
| CGS system | cm | g | s | Gal | dyn | $dyn/cm^2$ | $dyn/cm^2$ | erg |
| Gravitation system | m | $kgf \cdot s^2/m$ | s | $m/s^2$ | kgf | $kgf/m^2$ | $kgf/m^2$ | $kgf \cdot m$ |

Conversion to SI unit

| Quantity | Unit designation | Symbol | Conversion rate to SI | SI unit designation | Symbol |
|---|---|---|---|---|---|
| Angle | Degree | ° | $\pi/180$ | Radian | rad |
| | Minute | ′ | $\pi/10\,800$ | | |
| | Second | ″ (sec) | $\pi/648\,000$ | | |
| Length | Meter | m | 1 | Meter | m |
| | Micron | $\mu$ | $10^{-6}$ | | |
| | Angstrom | Å | $10^{-10}$ | | |
| Area | Square meter | $m^2$ | 1 | Square meter | $m^2$ |
| | Are | a | $10^2$ | | |
| | Hector | ha | $10^4$ | | |
| Volume | Cubic meter | $m^3$ | 1 | Cubic meter | $m^3$ |
| | Liter | R.L | $10^{-3}$ | | |
| Mass | kilogram | kg | 1 | Kilogram | kg |
| | Ton | t | $10^3$ | | |
| | Weight kilogram.square second per meter | $kgf \cdot s^2/m$ | 9.806 65 | | |
| Time | Second | s | 1 | Second | s |
| | Minute | min | 60 | | |
| | Hour | h | 3 600 | | |
| | Day | d | 86 400 | | |
| Speed | Meter per second | m/s | 1 | Meter per second | m/s |
| | Knot | kn | 1 852/3 600 | | |
| Frequency and vibration | Cycle | $s^{-1}$ (pps) | 1 | Hertz | Hz |
| Revolutions (rotational speed) | Revolutions per minute per second | rpm (r/min) | 1/60 | Per second | $s^{-1}$ |
| Angular speed | Radian per second | rad/s | 1 | Radian per second | rad/s |
| Acceleration | Meter per second per second | $m/s^2$ | 1 | Meter per second per second | $m/s^2$ |
| | G | G | 9.806 65 | | |
| Force | Weight kilogram | kgf | 9.806 65 | Newton | N |
| | Weight ton | tf | 9 806.65 | | |
| | Dyne | dyn | $10^{-5}$ | | |
| Force moment | Weight kilogram meter | $kgf \cdot m$ | 9.806 65 | Newton meter | $N \cdot m$ |
| Inertia moment | Weight kilogram per square meter | $kgf \cdot m \cdot s^2$ | 9.806 65 | kilogram square meter | $kg \cdot m^2$ |
| Stress | Weight kilogram per square meter | $kgf/m^2$ | 9.806 65 | Pascal or Newton per square meter | Pa or $N/m^2$ |
| Pressure | Weight kilogram per square meter | $kgf/m^2$ | 9.806 65 | Pascal | Pa |
| | Water column meter | $mH_2O$ | 9 806.65 | | |
| | Mercury column meter | mHg | 101 325/0.76 | | |
| | Torr | Torr | 101 325/760 | | |
| | Atmospheric pressure | atm | 101 325 | | |
| | Bar | bar | $10^5$ | | |
| Energy | Erg | erg | $10^{-7}$ | Joule | J |
| | IT calorie | $cal_{IT}$ | 4.186 8 | | |
| | Weight kilogram meter | $kgf \cdot m$ | 9.806 65 | | |
| | Kilowatt per hour | $kW \cdot h$ | $3.600 \times 10^6$ | | |
| | Horse power per hour | $PS \cdot h$ | $2.647\,79 \times 10^6$ | | |
| Power rate and power | Watt | W | 1 | Watt | W |
| | Horse power | PS | 735.5 | | |
| | Weight kilogram meter per second | $kgf \cdot m/s$ | 9.806 65 | | |

| Quantity / Unit system | Power rate | Temperature | Viscosity | Dynamic viscosity | Magnetic flux | Flux density | Magnetic field strength |
|---|---|---|---|---|---|---|---|
| SI | W | K | Pa · s | $m^2/s$ | Wb | T | A/m |
| CGS system | erg/s | ˚C | P | St | Mx | Gs | Oe |
| Gravitation system | kgf · m/s | ˚C | kgf · $s/m^2$ | $m^2/s$ | — | — | — |

Conversion to SI unit

| Quantity | Unit designation | Symbol | Conversion rate to SI | SI unit designation | Symbol |
|---|---|---|---|---|---|
| Viscosity | Poise | P | $10^{-1}$ | Pascal second | Pa · s |
| | Centi poise | cP | $10^{-3}$ | | |
| | Weight kilogram second per square meter | kgf · $s/m^2$ | 9.806 65 | | |
| Dynamic viscosity | Stokes | St | $10^{-4}$ | Square meter per second | $m^2/s$ |
| | Centistokes | cSt | $10^{-6}$ | | |
| Temperature | Degree | ˚C | +273.15 | Kelvin | K |
| Radioactive | Curie | Ci | $3.7\times10^{10}$ | Becquere | Bq |
| Dosage | Roentgen | R | $2.58\times10^{-4}$ | Coulomb per kilogram | C/kg |
| Absorption dosage | Rad | rad | $10^{-2}$ | Gray | Gy |
| Dosage equivalent | Rem | rem | $10^{-2}$ | Sivert | Sv |
| Magnetic flux | Maxwell | Mx | $10^{-8}$ | Weber | Wb |
| Flux density | Gamma | $\gamma$ | $10^{-9}$ | Tesler | T |
| | Gauss | Gs | $10^{-4}$ | | |
| Magnetic field strength | Oersted | Oe | $10^3/4\pi$ | Ampere per meter | A/m |
| Quantity of electricity | Coulomb | C | 1 | Coulomb | C |
| Potential difference | Volt | V | 1 | Volt | V |
| Electric resistance | Ohm | Ω | 1 | Ohm | Ω |
| Current | Ampere | A | 1 | Ampere | A |

Integer multiplication of SI unit 10

| Unit×multiple | Initial letter | |
|---|---|---|
| | Name | Symbol |
| $10^{18}$ | Exa | E |
| $10^{15}$ | Peta | P |
| $10^{12}$ | Tera | T |
| $10^{9}$ | Guiga | G |
| $10^{6}$ | Mega | M |
| $10^{3}$ | Kilo | k |
| $10^{2}$ | Hect | h |
| 10 | Daca | da |
| $10^{-1}$ | Deci | d |
| $10^{-2}$ | Centi | c |
| $10^{-3}$ | Mili | m |
| $10^{-6}$ | Micro | $\mu$ |
| $10^{-9}$ | Nano | n |
| $10^{-12}$ | Pico | p |
| $10^{-15}$ | Femt | f |
| $10^{-18}$ | Atto | a |

kgf-N conversion table

| kgf | | N | kgf | | N | kgf | | N |
|---|---|---|---|---|---|---|---|---|
| 0.1020 | 1 | 9.8066 | 3.4670 | 34 | 333.43 | 6.8321 | 67 | 657.04 |
| 0.2039 | 2 | 19.613 | 3.5690 | 35 | 343.23 | 6.9341 | 68 | 666.85 |
| 0.3059 | 3 | 29.420 | 3.6710 | 36 | 353.04 | 7.0361 | 69 | 676.66 |
| 0.4079 | 4 | 39.227 | 3.7730 | 37 | 362.85 | 7.1380 | 70 | 686.46 |
| 0.5099 | 5 | 49.033 | 3.8749 | 38 | 372.65 | 7.2400 | 71 | 696.27 |
| 0.6118 | 6 | 58.840 | 3.9769 | 39 | 382.46 | 7.3420 | 72 | 706.08 |
| 0.7138 | 7 | 68.646 | 4.0789 | 40 | 392.27 | 7.4440 | 73 | 715.88 |
| 0.8158 | 8 | 78.453 | 4.1808 | 41 | 402.07 | 7.5459 | 74 | 725.69 |
| 0.9177 | 9 | 88.260 | 4.2828 | 42 | 411.88 | 7.6479 | 75 | 735.50 |
| 1.0197 | 10 | 98.066 | 4.3848 | 43 | 421.68 | 7.7499 | 76 | 745.30 |
| 1.1217 | 11 | 107.87 | 4.4868 | 44 | 431.49 | 7.8518 | 77 | 755.11 |
| 1.2237 | 12 | 117.68 | 4.5887 | 45 | 441.30 | 7.9538 | 78 | 764.92 |
| 1.3256 | 13 | 127.49 | 4.6907 | 46 | 451.10 | 8.0558 | 79 | 774.72 |
| 1.4276 | 14 | 137.29 | 4.7927 | 47 | 460.91 | 8.1578 | 80 | 784.53 |
| 1.5296 | 15 | 147.10 | 4.8946 | 48 | 470.72 | 8.2597 | 81 | 794.34 |
| 1.6316 | 16 | 156.91 | 4.9966 | 49 | 480.52 | 8.3617 | 82 | 804.14 |
| 1.7335 | 17 | 166.71 | 5.0986 | 50 | 490.33 | 8.4637 | 83 | 813.95 |
| 1.8355 | 18 | 176.52 | 5.2006 | 51 | 500.14 | 8.5656 | 84 | 823.76 |
| 1.9375 | 19 | 186.33 | 5.3025 | 52 | 509.94 | 8.6676 | 85 | 833.56 |
| 2.0394 | 20 | 196.13 | 5.4045 | 53 | 519.75 | 8.7696 | 86 | 843.37 |
| 2.1414 | 21 | 205.94 | 5.5065 | 54 | 529.56 | 8.8716 | 87 | 853.18 |
| 2.2434 | 22 | 215.75 | 5.6085 | 55 | 539.36 | 8.9735 | 88 | 862.98 |
| 2.3454 | 23 | 225.55 | 5.7104 | 56 | 549.17 | 9.0755 | 89 | 872.79 |
| 2.4473 | 24 | 235.36 | 5.8124 | 57 | 558.98 | 9.1775 | 90 | 882.60 |
| 2.5493 | 25 | 245.17 | 5.9144 | 58 | 568.78 | 9.2794 | 91 | 892.40 |
| 2.6513 | 26 | 254.97 | 6.0163 | 59 | 578.59 | 9.3814 | 92 | 902.21 |
| 2.7532 | 27 | 264.78 | 6.1183 | 60 | 588.40 | 9.4834 | 93 | 912.02 |
| 2.8552 | 28 | 274.59 | 6.2203 | 61 | 598.20 | 9.5854 | 94 | 921.82 |
| 2.9572 | 29 | 284.39 | 6.3223 | 62 | 608.01 | 9.6873 | 95 | 931.63 |
| 3.0592 | 30 | 294.20 | 6.4242 | 63 | 617.82 | 9.7893 | 96 | 941.44 |
| 3.1611 | 31 | 304.01 | 6.5262 | 64 | 627.62 | 9.8913 | 97 | 951.24 |
| 3.2631 | 32 | 313.81 | 6.6282 | 65 | 637.43 | 9.9932 | 98 | 961.05 |
| 3.3651 | 33 | 323.62 | 6.7302 | 66 | 647.24 | 10.0952 | 99 | 970.86 |

[How to read the table]
For example, for conversion from 10kgf to N read 10 on center stage of 1st row box and then read the right-side N.
10kgf is equivalent to 98.066 upon reading. Furthermore, for conversion from 10N to kfg read the right side kgf box.
10N is equivalent to 1.0197kg upon reading.

1kgf＝9.80665N
1N＝0.101972kgf

Viscosity conversion table

| Dynamic viscosity mm²/s | Saybolt SUS (sec) | Redwood R" (sec) | Engler E (deg.) |
|---|---|---|---|
| 2.7 | 35 | 32.2 | 1.18 |
| 4.3 | 40 | 36.2 | 1.32 |
| 5.9 | 45 | 40.6 | 1.46 |
| 7.4 | 50 | 44.9 | 1.60 |
| 8.9 | 55 | 49.1 | 1.75 |
| 10.4 | 60 | 53.5 | 1.88 |
| 11.8 | 65 | 57.9 | 2.02 |
| 13.1 | 70 | 62.3 | 2.15 |
| 14.5 | 75 | 67.6 | 2.31 |
| 15.8 | 80 | 71.0 | 2.42 |
| 17.0 | 85 | 75.1 | 2.55 |
| 18.2 | 90 | 79.6 | 2.68 |
| 19.4 | 95 | 84.2 | 2.81 |
| 20.6 | 100 | 88.4 | 2.95 |
| 23.0 | 110 | 97.1 | 3.21 |
| 25.0 | 120 | 105.9 | 3.49 |
| 27.5 | 130 | 114.8 | 3.77 |
| 29.8 | 140 | 123.6 | 4.04 |
| 32.1 | 150 | 132.4 | 4.32 |
| 34.3 | 160 | 141.1 | 4.59 |
| 36.5 | 170 | 150.0 | 4.88 |
| 38.8 | 180 | 158.8 | 5.15 |
| 41.0 | 190 | 167.5 | 5.44 |
| 43.2 | 200 | 176.4 | 5.72 |
| 47.5 | 220 | 194.0 | 6.28 |
| 51.9 | 240 | 212 | 6.85 |
| 56.5 | 260 | 229 | 7.38 |
| 60.5 | 280 | 247 | 7.95 |
| 64.9 | 300 | 265 | 8.51 |
| 70.3 | 325 | 287 | 9.24 |
| 75.8 | 350 | 309 | 9.95 |
| 81.2 | 375 | 331 | 10.7 |
| 86.8 | 400 | 353 | 11.4 |
| 92.0 | 425 | 375 | 12.1 |
| 97.4 | 450 | 397 | 12.8 |

| Dynamic viscosity mm²/s | Saybolt SUS (sec) | Redwood R" (sec) | Engler E (deg.) |
|---|---|---|---|
| 103 | 475 | 419 | 13.5 |
| 108 | 500 | 441 | 14.2 |
| 119 | 550 | 485 | 15.6 |
| 130 | 600 | 529 | 17.0 |
| 141 | 650 | 573 | 18.5 |
| 152 | 700 | 617 | 19.9 |
| 163 | 750 | 661 | 21.3 |
| 173 | 800 | 705 | 22.7 |
| 184 | 850 | 749 | 24.2 |
| 195 | 900 | 793 | 25.6 |
| 206 | 950 | 837 | 27.0 |
| 217 | 1 000 | 882 | 28.4 |
| 260 | 1 200 | 1 058 | 34.1 |
| 302 | 1 400 | 1 234 | 39.8 |
| 347 | 1 600 | 1 411 | 45.5 |
| 390 | 1 800 | 1 587 | 51 |
| 433 | 2 000 | 1 763 | 57 |
| 542 | 2 500 | 2 204 | 71 |
| 650 | 3 000 | 2 646 | 85 |
| 758 | 3 500 | 3 087 | 99 |
| 867 | 4 000 | 3 526 | 114 |
| 974 | 4 500 | 3 967 | 128 |
| 1 082 | 5 000 | 4 408 | 142 |
| 1 150 | 5 500 | 4 849 | 156 |
| 1 300 | 6 000 | 5 290 | 170 |
| 1 400 | 6 500 | 5 730 | 185 |
| 1 510 | 7 000 | 6 171 | 199 |
| 1 630 | 7 500 | 6 612 | 213 |
| 1 740 | 8 000 | 7 053 | 227 |
| 1 850 | 8 500 | 7 494 | 242 |
| 1 960 | 9 000 | 7 934 | 256 |
| 2 070 | 9 500 | 8 375 | 270 |
| 2 200 | 10 000 | 8 816 | 284 |

## Ordinary tolerance for cutting dimensions

JIS B 0405

Ordinary tolerance is applied to cutting dimensions for which special accuracy is not required functionally and, therefore, applicable tolerances are indicated in batch, without individual indication of them, in relevant specification, drawings, etc.

Ordinary tolerances shall be indicated by either one of the following methods (1) and (2).

(1) Numerical value table for each dimensional division

(2) Applicable Standard No. and tolerance class

Ex. Ordinary tolerance shall conform to the "Medium Class" specified in JIS B 0405.

Unit : mm

| Dimensional division \ Class | Fine class (Class-12) | Medium class (Class-14) | Coarse class (Class-16) |
|---|---|---|---|
| 0.5 and over 3 incl. | ±0.05 | ±0.1 | – |
| Over 3 6 incl. | | | ±0.2 |
| Over 6 30 incl. | ±0.1 | ±0.2 | ±0.5 |
| Over 30 120 incl. | ±0.15 | ±0.3 | ±0.8 |
| Over 120 315 incl. | ±0.2 | ±0.5 | ±1.2 |
| Over 315 1000 incl. | ±0.3 | ±0.8 | ±2 |
| Over 1000 2000 incl. | ±0.5 | ±1.2 | ±3 |

Reference: The numerical values given above every each of “Fine Class”, “Medium class” and “Coarse Class” match “Fine series, Medium series”, and Coarse series” specified in ISO 2768 (Permissible Machining Variations in Dimensions without Tolerance

## Ordinary tolerances for castings

JIS B 0407

This Standard specifies the ordinary tolerances for the as-casted length and wall thickness dimensions (hereinafter referred to as" ordinary tolerance") of gray castings and spherical graphite

castings molded using sand molds (excluding precision mold and other equivalent).

The ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

(1) Numerical value table for each dimensional division

(2) Applicable Standard No. and tolerance class

Ex. JIS B 0407, Coarse class

Ordinary tolerance for length

Unit : mm

| Material / Class \ Dimensional division | Gray castings | | Spherical graphite castings | |
|---|---|---|---|---|
| | Fine class | Coarse class | Fine class | Coarse class |
| 120 incl. | ±1 | ± 1.5 | ±1.5 | ± 2 |
| Over 120 250 incl. | ±1.5 | ± 2 | ±2 | ± 2.5 |
| Over 250 400 incl. | ±2 | ± 3 | ±2.5 | ± 3.5 |
| Over 400 800 incl. | ±3 | ± 4 | ±4 | ± 5 |
| Over 800 1600 incl. | ±4 | ± 6 | ±5 | ± 7 |
| Over 1600 3150 incl. | – | ±10 | – | ±10 |

Ordinary tolerance for wall thickness

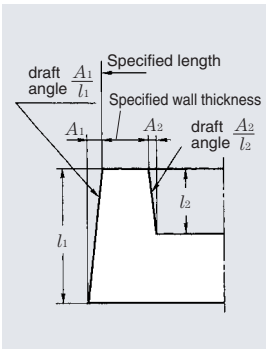
Unit : mm

| Material / Class \ Dimensional division | Gray castings | | Spherical graphite castings | |
|---|---|---|---|---|
| | Fine class | Coarse class | Fine class | Coarse class |
| 10 incl. | ±1 | ±1.5 | ±1.2 | ±2 |
| Over 10 18 incl. | ±1.5 | ±2 | ±1.5 | ±2.5 |
| Over 18 30 incl. | ±2 | ±3 | ±2 | ±3 |
| Over 30 50 incl. | ±2 | ±3.5 | ±2.5 | ±4 |

Dimension-A for draft angle

Unit : mm

| Dimensional division $l$ | Dimension $A$ (max) |
|---|---|
| 18 incl. | 1 |
| Over 18 30 incl. | 1.5 |
| Over 30 50 incl. | 2 |
| Over 50 120 incl. | 2.5 |
| Over 120 315 incl. | 3.5 |
| Over 315 630 incl. | 6 |
| Over 630 1000 incl. | 9 |




Remarks : $l$ means $l_1$ and $l_2$ in the above sketch.
$A$ means $A_1$, $A_2$ in the above sketch.

## Ordinary tolerances for press-formed metal parts

JIS B 0408

This Standard specifies the ordinary tolerances for the punching, bending and drawing dimensions, bending angle of press-formed metal parts (hereinafter referred to as "ordinary tolerance").

Ordinary tolerance is applied to the dimensions for which special accuracy is not required functionally and, therefore, applicable tolerances are indicated in batch, without individual indication of them, in relevant specification, drawings, etc.

Ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

(1) Numerical value table for each dimensional division

(2) Applicable Standard No. and tolerance class

Ex. JIS B0408, Class-A

Ordinary tolerance for punching, bending and drawing dimensions

Unit : mm

| Dimensional division \ Class | Class-A | Class-B | Class-C | Class-D | Class-E |
|---|---|---|---|---|---|
| 30 incl. | ±0.1 | ±0.15 | ±0.25 | ±0.4 | ±0.6 |
| Over 30 120 incl. | ±0.15 | ±0.25 | ±0.45 | ±0.7 | ±1.1 |
| Over 120 315 incl. | ±0.2 | ±0.4 | ±0.6 | ±1 | ±1.6 |
| Over 315 1000 incl. | ±0.3 | ±0.7 | ±1.1 | ±1.8 | ±2.8 |
| Over 1000 2000 incl. | ±0.5 | ±1.1 | ±1.8 | ±3 | ±4.5 |

Ordinary tolerance for bending angle

Unit : degree

| Type of bending \ Class | Class B, C | Class D, E |
|---|---|---|
| Perpendicular bending | ±1 | ±2 |
| Other bending | ±1.5 | ±3 |

Remarks: The ordinary tolerance for bending angle shall be applied to Class-A tolerance.

## Ordinary tolerance for die casts

JIS B 0409

This Standard specifies the ordinary tolerance for the as-casted dimensions of zinc allow die casts and aluminum alloy die casts.

The ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

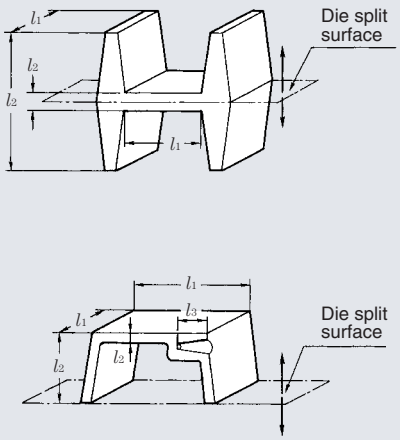
(1) Numerical value table for each dimensional division

(2) Applicable standard No. Ex.:JIS B 0409

Ordinary tolerance for dimensions

Unit : mm

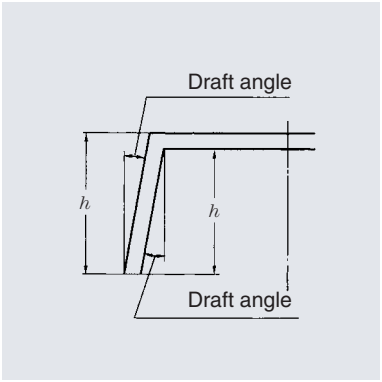
| Classification by application / Dimensional division | Portions casted by solid die and movable die | | | Portions casted by movable core $l_2$ | |
|---|---|---|---|---|---|
| | Parallel to die split surface | Perpendicular to die split surface[1] $l_2$ | | | |
| | | Projected area of castings perpendicular to die split surface[2] cm² | | Projected area of casting portion perpendicular to moving direction of movable core cm² | |
| | | 600 and below | over 600 2400 incl. | 150 and below | over 150 600 incl. |
| 30 incl. | ±0.25 | ±0.5 | ±0.6 | ±0.5 | ±0.6 |
| Over 30 50 incl. | ±0.3 | ±0.5 | ±0.6 | ±0.5 | ±0.6 |
| Over 50 80 incl. | ±0.35 | ±0.6 | ±0.6 | ±0.6 | ±0.6 |
| Over 80 120 incl. | ±0.45 | ±0.7 | ±0.7 | ±0.7 | ±0.7 |
| Over 120 180 incl. | ±0.5 | ±0.8 | ±0.8 | ±0.8 | ±0.8 |
| Over 180 250 incl. | ±0.55 | ±0.9 | ±0.9 | ±0.9 | ±0.9 |
| Over 250 315 incl. | ±0.6 | ±1 | ±1 | ±1 | ±1 |
| Over 315 400 incl. | ±0.7 | – | – | – | – |
| Over 400 500 incl. | ±0.8 | – | – | – | – |
| Over 500 630 incl. | ±0.9 | – | – | – | – |
| Over 630 800 incl. | ±1 | – | – | – | – |
| Over 800 1000 incl. | ±1.1 | – | – | – | – |




Note 1) The dimensional tolerance for $l_1$ shall be applied to partial dimension of die split surface which does not influence the length.

2) “Projected area of castings” means the circumference internal projected area of as-casted castings.

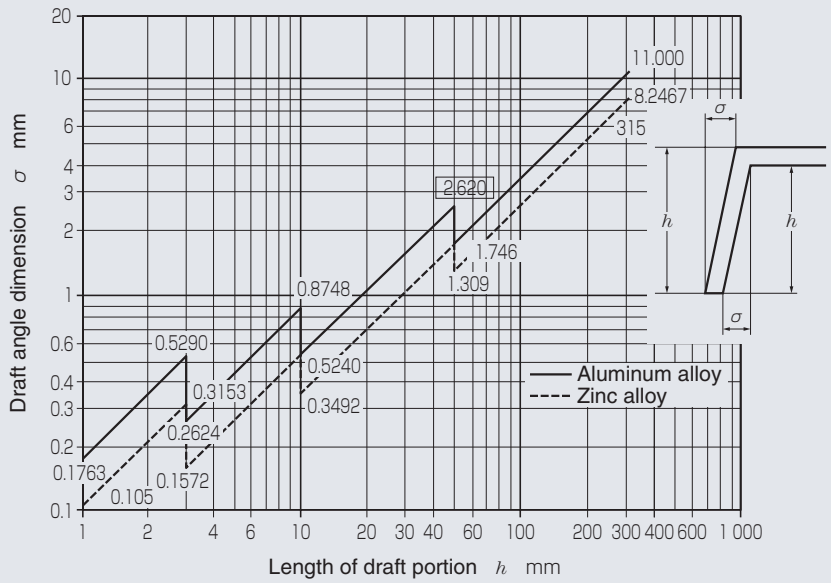
Maximum draft angle value




Unit : degree

| Material / Length of draft portion(mm) | Aluminum alloy | Zinc alloy |
|---|---|---|
| 3 and less | 10 | 6 |
| Over 3 - 10 incl. | 5 | 3 |
| Over 10 - 50 incl. | 3 | 2 |
| Over 50 - 315 incl. | 2 | 1.5 |

Chart showing the relationship of drafted portion length ($h$) vs draft angle dimension ($\sigma$)




## Ordinary tolerances for sheared metal plates

JIS B 0410

This Standard specifies the ordinary tolerances for the shearing width and ordinary tolerances for the straightness and perpendicularity (hereinafter generically referred to as "ordinary tolerance") of metal plates of 12mm and less in thickness which were sheared by direct shearing machines such as gear pusher, square shear, etc.

The ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

(1) Numerical value table for each dimensional division

(2) Applicable standard No. and tolerance class

Ex. 1. JIS B 0410, Class-B

2. Cutting width, straightness: JIS B 0410, Class-B
   Perpendicularity: JIS B 0410, Class-A

Definition of the terms: The terms used in this Standard are as defined in JIS B 0111 (Terminology relating to press machines) and, in addition, defined as follows.

(1) Cutting width: Distance from one side sheared by shear cutter to opposite side, as illustrated in Fig. $b$

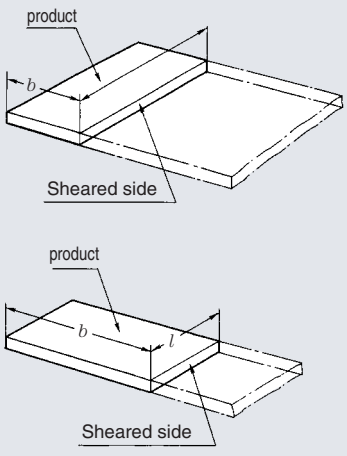
(2) Cutting length: length of one side sheared by shear cutter, as illustrated in Fig. $l$.

Ordinary tolerance for shearing width

Unit : mm

| Classification by plate thickness / Tolerance class / Cutting width division | 1.6 and less | | Over 1.6 3 incl. | | Over 3 6 incl. | | Over 6 12 incl. | |
|---|---|---|---|---|---|---|---|---|
| | Class-A | Class-B | Class-A | Class-B | Class-A | Class-B | Class-A | Class-B |
| 120 and less | ±0.2 | ±0.5 | ±0.3 | ±0.5 | ±0.7 | ±1.2 | – | ±1.6 |
| Over 120 - 315 incl. | ±0.3 | ±0.6 | ±0.4 | ±0.8 | ±0.9 | ±1.6 | – | ±2.0 |
| Over 315 - 1000 incl. | ±0.5 | ±1.0 | ±0.5 | ±1.2 | ±1.6 | ±2.0 | – | ±2.5 |
| Over 1000 - 2000 incl. | ±0.8 | ±1.2 | ±0.8 | ±1.6 | ±2.0 | ±3.0 | – | ±3.0 |
| Over 2000 - 3150 incl. | ±1.0 | ±1.4 | ±1.0 | ±1.6 | ±2.8 | ±3.0 | – | ±3.5 |

Remarks: non-applicable to cutting length exceeding 3150 mm.



Ordinary tolerance for straightness

Unit : mm

| Classification by plate thickness / Tolerance class / Cutting width division | 1.6 and less | | Over 1.6 3 incl. | | Over 3 6 incl. | |
|---|---|---|---|---|---|---|
| | Class-A | Class-B | Class-A | Class-B | Class-A | Class-B |
| 120 and less | 0.2 | 0.3 | 0.2 | 0.3 | 0.5 | 0.8 |
| Over 120 - 315 incl. | 0.3 | 0.5 | 0.3 | 0.6 | 1.0 | 1.6 |
| Over 315 - 1000 incl. | 0.4 | 0.9 | 0.5 | 1.0 | 1.4 | 2.0 |
| Over 1000 - 2000 incl. | 0.6 | 1.2 | 0.8 | 1.6 | 2.5 | 3.0 |
| Over 2000 - 3150 incl. | 0.9 | 1.6 | 1.0 | 2.2 | 2.8 | 4.5 |

Remarks: Non-applicable to cutting width of less than 20 times as much as plate thickness or cutting width of less than 30mm.

Ordinary tolerance for straightness

Unit : mm

| Classification by plate thickness / Tolerance class / Cutting width division | 3 and less | | Over 3 6 incl. | | Over 6 12 incl. | |
|---|---|---|---|---|---|---|
| | Class-A | Class-B | Class-A | Class-B | Class-A | Class-B |
| 120 and less | 0.3 | 0.5 | 0.5 | 1.0 | — | 1.6 |
| Over 120 - 315 incl. | 0.6 | 1.2 | 0.8 | 1.6 | — | 2.0 |
| Over 315 - 1000 incl. | 1.6 | 3.0 | 2.0 | 3.0 | — | 3.0 |
| Over 1000 - 2000 incl. | 3.0 | 6.0 | 4.0 | 6.0 | — | 6.0 |
| Over 2000 - 3150 incl. | 4.5 | 9.0 | 5.5 | 9.0 | — | 9.0 |

## Ordinary tolerance for sintered metal parts

JIS B 0411

This Standard specifies the ordinary tolerance which is applied to the machining dimensions of sintered mechanical parts and sintered oil-contained bearing, of sintered metal parts, but excluding machining methods other (e.g. cutting, etc.) than the machining method specific for sintered metal parts.

Ordinary tolerance is applied to the dimensions for which special accuracy is not required functionally and, therefore, applicable tolerances are indicated in batch, without individual indication of them, in relevant specification, drawings, etc.

Ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

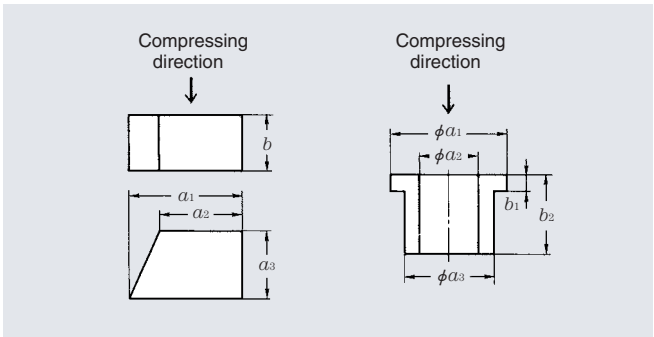
(1) Numerical value table for each dimensional division

(2) Applicable Standard No. and tolerance class
Ex. JIS B0411, Fine Class

Definition of terms: The terms used in this Standard are as defined below.

(1) Width: Perpendicular dimension against compressing direction in compressive molding of powders, as illustrated in Fig. $a$.

(2) Height: Dimension parallel to compressing direction in molding of powders, as illustrated in Fig. $b$.




Class: The ordinary tolerance class shall be three classes of “Fine Class”, “Medium Class” and “Coarse class”.

Reference: The numerical values every each class of “Fine Class”, “Medium Class” and “Coarse class” match “Fine Series, Medium Series and Coarse Series specified in ISO 2768 (Permissible machining variations in dimensions without tolerance indication)

Ordinary tolerance for width

Unit : mm

| class / Dimensional division | Fine class | Medium class | Coarse class |
|---|---|---|---|
| 6 and less | ±0.05 | ±0.1 | ±0.2 |
| Over 6 - 30 incl. | ±0.1 | ±0.2 | ±0.5 |
| Over 30 - 120 incl. | ±0.15 | ±0.3 | ±0.8 |
| Over 120 - 315 incl. | ±0.2 | ±0.5 | ±1.2 |

Ordinary tolerance for height

Unit : mm

| class / Dimensional division | Fine class | Medium class | Coarse class |
|---|---|---|---|
| 6 and less | ±0.1 | ±0.2 | ±0.6 |
| Over 6 - 30 incl. | ±0.2 | ±0.5 | ±1 |
| Over 30 - 120 incl. | ±0.3 | ±0.8 | ±1.8 |

## Ordinary tolerance for steel castings

JIS B 0412

This Standard specifies the ordinary tolerances for the length and as-casted wall thickness (hereinafter referred to as "ordinary tolerance") of steel castings casted by sand molds.

Ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

(1) Numerical value table for each dimensional division

(2) Applicable Standard No. and tolerance class
Ex. JIS B0412, Medium Class

Tolerance class: The ordinary tolerance class shall be three classes of "Fine Class", "Medium Class" and "Coarse Class".

Ordinary tolerance for length

Unit : mm

| Dimensional division \ class | Fine class | Medium class | Coarse class |
|---|---|---|---|
| 120 and less | ±1.8 | ± 2.8 | ± 4.5 |
| Over 120 - 315 incl. | ±2.5 | ± 4 | ± 6 |
| Over 315 - 630 incl. | ±3.5 | ± 5.5 | ± 9 |
| Over 630 - 1250 incl. | ±5 | ± 8 | ±12 |
| Over 1250 - 2500 incl. | ±9 | ±14 | ±22 |
| Over 2500 - 5000 incl. | – | ±20 | ±35 |
| Over 5000 - 10000 incl. | – | – | ±63 |

Ordinary tolerance for wall thickness

Unit : mm

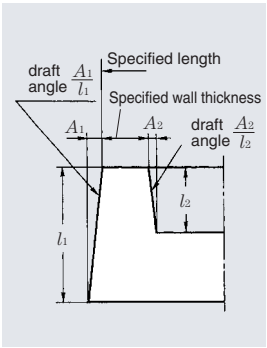
| Dimensional division \ class | Fine class | Medium class | Coarse class |
|---|---|---|---|
| 18 and less | ±1.4 | ±2.2 | ± 3.5 |
| Over 18 - 50 incl. | ±2 | ±3 | ± 5 |
| Over 50 - 120 incl. | – | ±4.5 | ± 7 |
| Over 120 - 250 incl. | – | ±5.5 | ± 9 |
| Over 250 - 400 incl. | – | ±7 | ±11 |
| Over 400 - 630 incl. | – | ±9 | ±14 |
| Over 630 - 1000 incl. | – | – | ±18 |

Remarks: “Fine Class” shall be applied to minor steel castings subjected to special accuracy.

Dimension-A for draft angle

Unit : mm

| Dimensional division $l$ | Dimension $A$ (max) |
|---|---|
| 18 incl. | 1.4 |
| Over 18 50 incl. | 2 |
| Over 50 120 incl. | 2.8 |
| Over 120 250 incl. | 3.5 |
| Over 250 400 incl. | 4.5 |
| Over 400 630 incl. | 5.5 |
| Over 630 1000 incl. | 7 |




Remarks: $l$ means $l_1$ and $l_2$ in the above sketch.
$A$ means $A_1$, $A_2$ in the above sketch.

## Ordinary tolerance for aluminum alloy castings

JIS B 0414

This Standard specifies the ordinary tolerances (hereinafter referred to as "ordinary tolerance") for the length and wall thickness of aluminum alloy castings (hereinafter referred to as "castings" molded by sand mold and mold. However, this Standard is not applied to castings molded by precision molds using "Lost Wax Method", etc.

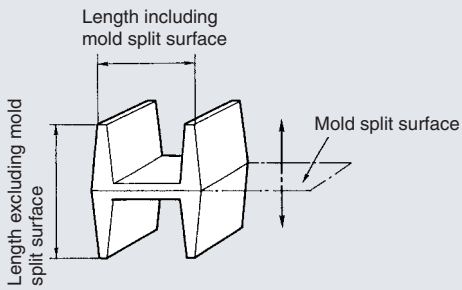
Ordinary tolerance shall be indicated by either one of the following methods (1) and (2).

(1) Numerical value table for each dimensional division

(2) Applicable Standard No. , classification and tolerance class by casting mold
Ex. JIS 0414 , Sand mold, Coarse Class

Definition of terms: The terms used in this Standard are as defined below.

Mold split surface: Two or more splitted surfaces of mold for drafting a model or a castings out of a casting mold.



Ordinary tolerance for length

Unit : mm

| Type | Dimensional division | 50 incl. | | Over 50 120 incl. | | Over 120 250 incl. | | Over 250 400 incl. | | Over 400 800 incl. | | Over 800 1600 incl. | | Over 1600 3150 incl. | | (Ref.) 1) Correspond to standard tolerance grade | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class |
| Castings by sand mold | Portion including no mold split surface | ±0.5 | ±1.1 | ±0.7 | ±1.2 | ±0.9 | ±1.4 | ±1.1 | ±1.8 | ±1.6 | ±2.5 | – | ±4 | – | ±7 | 15 | 16 |
| | Portion including mold split surface | ±0.8 | ±1.5 | ±1.1 | ±1.8 | ±1.4 | ±2.2 | ±1.8 | ±2.8 | ±2.5 | ±4.0 | – | ±6 | – | – | 16 | 17 |
| Castings by casting mold | Portion including no mold split surface | ±0.3 | ±0.5 | ±0.45 | ±0.7 | ±0.55 | ±0.9 | ±0.7 | ±1.1 | ±1.0 | ±1.6 | – | – | – | – | 14 | 15 |
| | Portion including mold split surface | ±0.5 | ±0.6 | ±0.7 | ±0.8 | ±0.9 | ±1.0 | ±1.1 | ±1.2 | ±1.6 | ±1.8 | – | – | – | – | 15 | 15 |

Note 1) Conforming to Appendix-2 to JIS B 0404 (Rule for Ordinary Dimensional Tolerances)

Ordinary tolerance for wall thickness

Unit : mm

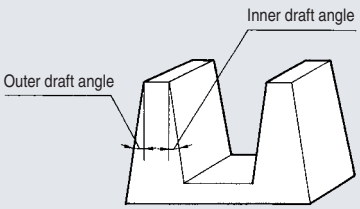
| Type | Max. length of castings \ Tolerance class \ Dimensional division | 6 incl. | | Over 6 10 incl. | | Over 10 18 incl. | | Over 18 30 incl. | | Over 30 50 incl. | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class | Fine class | Coarse class |
| Castings by sand mold | 120 incl. | ±0.6 | ±1.2 | ±0.7 | ±1.4 | ±0.8 | ±1.6 | ±0.9 | ±1.8 | – | – |
| | Over 120 250 incl. | ±0.7 | ±1.3 | ±0.8 | ±1.5 | ±0.9 | ±1.7 | ±1.0 | ±1.9 | ±1.2 | ±2.3 |
| | Over 250 400 incl. | ±0.8 | ±1.4 | ±0.9 | ±1.6 | ±1.0 | ±1.8 | ±1.1 | ±2.0 | ±1.3 | ±2.4 |
| | Over 400 800 incl. | ±1.0 | ±1.6 | ±1.1 | ±1.8 | ±1.2 | ±2.0 | ±1.3 | ±2.2 | ±1.5 | ±2.6 |
| Castings by casting mold | 120 incl. | ±0.3 | ±0.7 | ±0.4 | ±0.9 | ±0.5 | ±1.1 | ±0.6 | ±1.3 | – | – |
| | Over 120 250 incl. | ±0.4 | ±0.8 | ±0.5 | ±1.0 | ±0.6 | ±1.2 | ±0.7 | ±1.4 | ±0.9 | ±1.8 |
| | Over 250 400 incl. | ±0.5 | ±0.9 | ±0.6 | ±1.1 | ±0.7 | ±1.3 | ±0.8 | ±1.5 | ±1.0 | ±1.9 |

Maximum draft angle value

Unit : degree

| Classification by draft angle | Outer | | Inner | |
|---|---|---|---|---|
| Tolerance class / Type of castings | Fine class | Coarse class | Fine class | Coarse class |
| Fine class<br>Coarse class | 1 | 2 | 2 | 3 |

Remarks: The numerical values in this table shall be applied to draft portions of 400 mm and less in length.



Dimensional tolerance for shaft

| Diameter division mm | | a13 | | c12 | | d6 | | e6 | | e13 | | f5 | | f6 | | g5 | | g6 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Over | incl. | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low |
| 3 | 6 | − 270 | − 450 | − 70 | − 190 | − 30 | − 38 | − 20 | − 28 | − 20 | − 200 | −10 | −15 | − 10 | − 18 | − 4 | − 9 | − 4 | − 12 |
| 6 | 10 | − 280 | − 500 | − 80 | − 230 | − 40 | − 49 | − 25 | − 34 | − 25 | − 245 | −13 | −19 | − 13 | − 22 | − 5 | −11 | − 5 | − 14 |
| 10 | 18 | − 290 | − 560 | − 95 | − 275 | − 50 | − 61 | − 32 | − 43 | − 32 | − 302 | −16 | −24 | − 16 | − 27 | − 6 | −14 | − 6 | − 17 |
| 18 | 30 | − 300 | − 630 | −110 | − 320 | − 65 | − 78 | − 40 | − 53 | − 40 | − 370 | −20 | −29 | − 20 | − 33 | − 7 | −16 | − 7 | − 20 |
| 30 | 40 | − 310 | − 700 | −120 | − 370 | − 80 | − 96 | − 50 | − 66 | − 50 | − 440 | −25 | −36 | − 25 | − 41 | − 9 | −20 | − 9 | − 25 |
| 40 | 50 | − 320 | − 710 | −130 | − 380 | | | | | | | | | | | | | | |
| 50 | 65 | − 340 | − 800 | −140 | − 440 | −100 | −119 | − 60 | − 79 | − 60 | − 520 | −30 | −43 | − 30 | − 49 | −10 | −23 | −10 | − 29 |
| 65 | 80 | − 360 | − 820 | −150 | − 450 | | | | | | | | | | | | | | |
| 80 | 100 | − 380 | − 920 | −170 | − 520 | −120 | −142 | − 72 | − 94 | − 72 | − 612 | −36 | −51 | − 36 | − 58 | −12 | −27 | −12 | − 34 |
| 100 | 120 | − 410 | − 950 | −180 | − 530 | | | | | | | | | | | | | | |
| 120 | 140 | − 460 | −1 090 | −200 | − 600 | −145 | −170 | − 85 | −110 | − 85 | − 715 | −43 | −61 | − 43 | − 68 | −14 | −32 | −14 | − 39 |
| 140 | 160 | − 520 | −1 150 | −210 | − 610 | | | | | | | | | | | | | | |
| 160 | 180 | − 580 | −1 210 | −230 | − 630 | | | | | | | | | | | | | | |
| 180 | 200 | − 660 | −1 380 | −240 | − 700 | −170 | −199 | −100 | −129 | −100 | − 820 | −50 | −70 | − 50 | − 79 | −15 | −35 | −15 | − 44 |
| 200 | 225 | − 740 | −1 460 | −260 | − 720 | | | | | | | | | | | | | | |
| 225 | 250 | − 820 | −1 540 | −280 | − 740 | | | | | | | | | | | | | | |
| 250 | 280 | − 920 | −1 730 | −300 | − 820 | −190 | −222 | −110 | −142 | −110 | − 920 | −56 | −79 | − 56 | − 88 | −17 | −40 | −17 | − 49 |
| 280 | 315 | −1 050 | −1 860 | −330 | − 850 | | | | | | | | | | | | | | |
| 315 | 355 | −1 200 | −2 090 | −360 | − 930 | −210 | −246 | −125 | −161 | −125 | −1 015 | −62 | −87 | − 62 | − 98 | −18 | −43 | −18 | − 54 |
| 355 | 400 | −1 350 | −2 240 | −400 | − 970 | | | | | | | | | | | | | | |
| 400 | 450 | −1 500 | −2 470 | −440 | −1 070 | −230 | −270 | −135 | −175 | −135 | −1 105 | −68 | −95 | − 68 | −108 | −20 | −47 | −20 | − 60 |
| 450 | 500 | −1 650 | −2 620 | −480 | −1 110 | | | | | | | | | | | | | | |
| 500 | 560 | − | − | − | − | −260 | −304 | −145 | −189 | − | − | − | − | − 76 | −120 | − | − | −22 | − 66 |
| 560 | 630 | | | | | | | | | | | | | | | | | | |
| 630 | 710 | − | − | − | − | −290 | −340 | −160 | −210 | − | − | − | − | − 80 | −130 | − | − | −24 | − 74 |
| 710 | 800 | | | | | | | | | | | | | | | | | | |
| 800 | 900 | − | − | − | − | −320 | −376 | −170 | −226 | − | − | − | − | − 86 | −142 | − | − | −26 | − 82 |
| 900 | 1 000 | | | | | | | | | | | | | | | | | | |
| 1 000 | 1 120 | − | − | − | − | −350 | −416 | −195 | −261 | − | − | − | − | − 98 | −164 | − | − | −28 | − 94 |
| 1 120 | 1 250 | | | | | | | | | | | | | | | | | | |
| 1 250 | 1 400 | − | − | − | − | −390 | −468 | −220 | −298 | − | − | − | − | −110 | −188 | − | − | −30 | −108 |
| 1 400 | 1 600 | | | | | | | | | | | | | | | | | | |

| Diameter division mm | | j5 | | js5 | | j6 | | js6 | | j7 | | k4 | | k5 | | k6 | | m5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Over | incl. | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low |
| 3 | 6 | +3 | − 2 | + 2.5 | − 2.5 | + 6 | − 2 | + 4 | − 4 | + 8 | − 4 | + 5 | +1 | + 6 | +1 | + 9 | +1 | + 9 | + 4 |
| 6 | 10 | +4 | − 2 | + 3 | − 3 | + 7 | − 2 | + 4.5 | − 4.5 | +10 | − 5 | + 5 | +1 | + 7 | +1 | +10 | +1 | +12 | + 6 |
| 10 | 18 | +5 | − 3 | + 4 | − 4 | + 8 | − 3 | + 5.5 | − 5.5 | +12 | − 6 | + 6 | +1 | + 9 | +1 | +12 | +1 | +15 | + 7 |
| 18 | 30 | +5 | − 4 | + 4.5 | − 4.5 | + 9 | − 4 | + 6.5 | − 6.5 | +13 | − 8 | + 8 | +2 | +11 | +2 | +15 | +2 | +17 | + 8 |
| 30 | 40 | +6 | − 5 | + 5.5 | − 5.5 | +11 | − 5 | + 8 | − 8 | +15 | −10 | + 9 | +2 | +13 | +2 | +18 | +2 | +20 | + 9 |
| 40 | 50 | | | | | | | | | | | | | | | | | | |
| 50 | 65 | +6 | − 7 | + 6.5 | − 6.5 | +12 | − 7 | + 9.5 | − 9.5 | +18 | −12 | +10 | +2 | +15 | +2 | +21 | +2 | +24 | +11 |
| 65 | 80 | | | | | | | | | | | | | | | | | | |
| 80 | 100 | +6 | − 9 | + 7.5 | − 7.5 | +13 | − 9 | +11 | −11 | +20 | −15 | +13 | +3 | +18 | +3 | +25 | +3 | +28 | +13 |
| 100 | 120 | | | | | | | | | | | | | | | | | | |
| 120 | 140 | +7 | −11 | + 9 | − 9 | +14 | −11 | +12.5 | −12.5 | +22 | −18 | +15 | +3 | +21 | +3 | +28 | +3 | +33 | +15 |
| 140 | 160 | | | | | | | | | | | | | | | | | | |
| 160 | 180 | | | | | | | | | | | | | | | | | | |
| 180 | 200 | +7 | −13 | +10 | −10 | +16 | −13 | +14.5 | −14.5 | +25 | −21 | +18 | +4 | +24 | +4 | +33 | +4 | +37 | +17 |
| 200 | 225 | | | | | | | | | | | | | | | | | | |
| 225 | 250 | | | | | | | | | | | | | | | | | | |
| 250 | 280 | +7 | −16 | +11.5 | −11.5 | +16 | −16 | +16 | −16 | +26 | −26 | +20 | +4 | +27 | +4 | +36 | +4 | +43 | +20 |
| 280 | 315 | | | | | | | | | | | | | | | | | | |
| 315 | 355 | +7 | −18 | +12.5 | −12.5 | +18 | −18 | +18 | −18 | +29 | −28 | +22 | +4 | +29 | +4 | +40 | +4 | +46 | +21 |
| 355 | 400 | | | | | | | | | | | | | | | | | | |
| 400 | 450 | +7 | −20 | +13.5 | −13.5 | +20 | −20 | +20 | −20 | +31 | −32 | +25 | +5 | +32 | +5 | +45 | +5 | +50 | +23 |
| 450 | 500 | | | | | | | | | | | | | | | | | | |
| 500 | 560 | − | − | − | − | − | − | +22 | −22 | − | − | − | − | − | − | +44 | 0 | − | − |
| 560 | 630 | | | | | | | | | | | | | | | | | | |
| 630 | 710 | − | − | − | − | − | − | +25 | −25 | − | − | − | − | − | − | +50 | 0 | − | − |
| 710 | 800 | | | | | | | | | | | | | | | | | | |
| 800 | 900 | − | − | − | − | − | − | +28 | −28 | − | − | − | − | − | − | +56 | 0 | − | − |
| 900 | 1 000 | | | | | | | | | | | | | | | | | | |
| 1 000 | 1 120 | − | − | − | − | − | − | +33 | −33 | − | − | − | − | − | − | +66 | 0 | − | − |
| 1 120 | 1 250 | | | | | | | | | | | | | | | | | | |
| 1 250 | 1 400 | − | − | − | − | − | − | +39 | −39 | − | − | − | − | − | − | +78 | 0 | − | − |
| 1 400 | 1 600 | | | | | | | | | | | | | | | | | | |

Unit : μm

| h4 | | h5 | | h6 | | h7 | | h8 | | h9 | | h10 | | h11 | | h13 | | js4 | | Diameter division mm | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | Over | incl. |
| 0 | − 4 | 0 | − 5 | 0 | − 8 | 0 | − 12 | 0 | − 18 | 0 | − 30 | 0 | − 48 | 0 | − 75 | 0 | −180 | + 2 | − 2 | 3 | 6 |
| 0 | − 4 | 0 | − 6 | 0 | − 9 | 0 | − 15 | 0 | − 22 | 0 | − 36 | 0 | − 58 | 0 | − 90 | 0 | −220 | + 2 | − 2 | 6 | 10 |
| 0 | − 5 | 0 | − 8 | 0 | −11 | 0 | − 18 | 0 | − 27 | 0 | − 43 | 0 | − 70 | 0 | −110 | 0 | −270 | + 2.5 | − 2.5 | 10 | 18 |
| 0 | − 6 | 0 | − 9 | 0 | −13 | 0 | − 21 | 0 | − 33 | 0 | − 52 | 0 | − 84 | 0 | −130 | 0 | −330 | + 3 | − 3 | 18 | 30 |
| 0 | − 7 | 0 | − 11 | 0 | −16 | 0 | − 25 | 0 | − 39 | 0 | − 62 | 0 | −100 | 0 | −160 | 0 | −390 | + 3.5 | − 3.5 | 30<br>40 | 40<br>50 |
| 0 | − 8 | 0 | −13 | 0 | −19 | 0 | − 30 | 0 | − 46 | 0 | − 74 | 0 | −120 | 0 | −190 | 0 | −460 | + 4 | − 4 | 50<br>65 | 65<br>80 |
| 0 | −10 | 0 | −15 | 0 | −22 | 0 | − 35 | 0 | − 54 | 0 | − 87 | 0 | −140 | 0 | −220 | 0 | −540 | + 5 | − 5 | 80<br>100 | 100<br>120 |
| 0 | −12 | 0 | −18 | 0 | −25 | 0 | − 40 | 0 | − 63 | 0 | −100 | 0 | −160 | 0 | −250 | 0 | −630 | + 6 | − 6 | 120<br>140<br>160 | 140<br>160<br>180 |
| 0 | −14 | 0 | −20 | 0 | −29 | 0 | − 46 | 0 | − 72 | 0 | −115 | 0 | −185 | 0 | −290 | 0 | −720 | + 7 | − 7 | 180<br>200<br>225 | 200<br>225<br>250 |
| 0 | −16 | 0 | −23 | 0 | −32 | 0 | − 52 | 0 | − 81 | 0 | −130 | 0 | −210 | 0 | −320 | 0 | −810 | + 8 | − 8 | 250<br>280 | 280<br>315 |
| 0 | −18 | 0 | −25 | 0 | −36 | 0 | − 57 | 0 | − 89 | 0 | −140 | 0 | −230 | 0 | −360 | 0 | −890 | + 9 | − 9 | 315<br>355 | 355<br>400 |
| 0 | −20 | 0 | −27 | 0 | −40 | 0 | − 63 | 0 | − 97 | 0 | −155 | 0 | −250 | 0 | −400 | 0 | −970 | +10 | −10 | 400<br>450 | 450<br>500 |
| − | − | − | − | 0 | −44 | 0 | − 70 | 0 | −110 | 0 | −175 | 0 | −280 | 0 | −440 | 0 | − | − | − | 500<br>560 | 560<br>630 |
| − | − | − | − | 0 | −50 | 0 | − 80 | 0 | −125 | 0 | −200 | 0 | −320 | 0 | −500 | 0 | − | − | − | 630<br>710 | 710<br>800 |
| − | − | − | − | 0 | −56 | 0 | − 90 | 0 | −140 | 0 | −230 | 0 | −360 | 0 | −560 | 0 | − | − | − | 800<br>900 | 900<br>1 000 |
| − | − | − | − | 0 | −66 | 0 | −105 | 0 | −165 | 0 | −260 | 0 | −420 | 0 | −660 | 0 | − | − | − | 1 000<br>1 120 | 1 120<br>1 250 |
| − | − | − | − | 0 | −78 | 0 | −125 | 0 | −195 | 0 | −310 | 0 | −500 | 0 | −780 | 0 | − | − | − | 1 250<br>1 400 | 1 400<br>1 600 |

Unit : μm

| m6 | | n5 | | n6 | | p5 | | p6 | | r6 | | r7 | | Basic tolerance | | | | Diameter division mm | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | IT2 | IT3 | IT5 | IT7 | Over | incl. |
| + 12 | + 4 | +13 | + 8 | + 16 | + 8 | +17 | +12 | + 20 | + 12 | + 23 | + 15 | + 27 | + 15 | 1.5 | 2.5 | 5 | 12 | 3 | 6 |
| + 15 | + 6 | +16 | +10 | + 19 | +10 | +21 | +15 | + 24 | + 15 | + 28 | + 19 | + 34 | + 19 | 1.5 | 2.5 | 6 | 15 | 6 | 10 |
| + 18 | + 7 | +20 | +12 | + 23 | +12 | +26 | +18 | + 29 | + 18 | + 34 | + 23 | + 41 | + 23 | 2 | 3 | 8 | 18 | 10 | 18 |
| + 21 | + 8 | +24 | +15 | + 28 | +15 | +31 | +22 | + 35 | + 22 | + 41 | + 28 | + 49 | + 28 | 2.5 | 4 | 9 | 21 | 18 | 30 |
| + 25 | + 9 | +28 | +17 | + 33 | +17 | +37 | +26 | + 42 | + 26 | + 50 | + 34 | + 59 | + 34 | 2.5 | 4 | 11 | 25 | 30<br>40 | 40<br>50 |
| + 30 | +11 | +33 | +20 | + 39 | +20 | +45 | +32 | + 51 | + 32 | + 60<br>+ 62 | + 41<br>+ 43 | + 71<br>+ 73 | + 41<br>+ 43 | 3 | 5 | 13 | 30 | 50<br>65 | 65<br>80 |
| + 35 | +13 | +38 | +23 | + 45 | +23 | +52 | +37 | + 59 | + 37 | + 73<br>+ 76 | + 51<br>+ 54 | + 86<br>+ 89 | + 51<br>+ 54 | 4 | 6 | 15 | 35 | 80<br>100 | 100<br>120 |
| + 40 | +15 | +45 | +27 | + 52 | +27 | +61 | +43 | + 68 | + 43 | + 88<br>+ 90<br>+ 93 | + 63<br>+ 65<br>+ 68 | +103<br>+105<br>+108 | + 63<br>+ 65<br>+ 68 | 5 | 8 | 18 | 40 | 120<br>140<br>160 | 140<br>160<br>180 |
| + 46 | +17 | +51 | +31 | + 60 | +31 | +70 | +50 | + 79 | + 50 | +106<br>+109<br>+113 | + 77<br>+ 80<br>+ 84 | +123<br>+126<br>+130 | + 77<br>+ 80<br>+ 84 | 7 | 10 | 20 | 46 | 180<br>200<br>225 | 200<br>225<br>250 |
| + 52 | +20 | +57 | +34 | + 66 | +34 | +79 | +56 | + 88 | + 56 | +126<br>+130 | + 94<br>+ 98 | +146<br>+150 | + 94<br>+ 98 | 8 | 12 | 23 | 52 | 250<br>280 | 280<br>315 |
| + 57 | +21 | +62 | +37 | + 73 | +37 | +87 | +62 | + 98 | + 62 | +144<br>+150 | +108<br>+114 | +165<br>+171 | +108<br>+114 | 9 | 13 | 25 | 57 | 315<br>355 | 355<br>400 |
| + 63 | +23 | +67 | +40 | + 80 | +40 | +95 | +68 | +108 | + 68 | +166<br>+172 | +126<br>+132 | +189<br>+195 | +126<br>+132 | 10 | 15 | 27 | 63 | 400<br>450 | 450<br>500 |
| + 70 | +26 | − | − | + 88 | +44 | − | − | +122 | + 78 | +194<br>+199 | +150<br>+155 | +220<br>+225 | +150<br>+155 | − | − | − | 70 | 500<br>560 | 560<br>630 |
| + 80 | +30 | − | − | +100 | +50 | − | − | +138 | + 88 | +225<br>+235 | +175<br>+185 | +255<br>+265 | +175<br>+185 | − | − | − | 80 | 630<br>710 | 710<br>800 |
| + 90 | +34 | − | − | +112 | +56 | − | − | +156 | +100 | +266<br>+276 | +210<br>+220 | +300<br>+310 | +210<br>+220 | − | − | − | 90 | 800<br>900 | 900<br>1 000 |
| +106 | +40 | − | − | +132 | +66 | − | − | +186 | +120 | +316<br>+326 | +250<br>+260 | +355<br>+365 | +250<br>+260 | − | − | − | 105 | 1 000<br>1 120 | 1 120<br>1 250 |
| +126 | +48 | − | − | +156 | +78 | − | − | +218 | +140 | +378<br>+408 | +300<br>+330 | +425<br>+455 | +300<br>+330 | − | − | − | 125 | 1 250<br>1 400 | 1 400<br>1 600 |

Dimensional tolerance for housing bore

| Diameter division mm Over | incl. | E7 High | E7 Low | E10 High | E10 Low | E11 High | E11 Low | E12 High | E12 Low | F6 High | F6 Low | F7 High | F7 Low | F8 High | F8 Low | G6 High | G6 Low | G7 High | G7 Low | H6 High | H6 Low |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 6 | + 32 | + 20 | + 68 | + 20 | + 95 | + 20 | +140 | + 20 | + 18 | + 10 | + 22 | + 10 | + 28 | + 10 | + 12 | + 4 | + 16 | + 4 | + 8 | 0 |
| 6 | 10 | + 40 | + 25 | + 83 | + 25 | +115 | + 25 | +175 | + 25 | + 22 | + 13 | + 28 | + 13 | + 35 | + 13 | + 14 | + 5 | + 20 | + 5 | + 9 | 0 |
| 10 | 18 | + 50 | + 32 | +102 | + 32 | +142 | + 32 | +212 | + 32 | + 27 | + 16 | + 34 | + 16 | + 43 | + 16 | + 17 | + 6 | + 24 | + 6 | +11 | 0 |
| 18 | 30 | + 61 | + 40 | +124 | + 40 | +170 | + 40 | +250 | + 40 | + 33 | + 20 | + 41 | + 20 | + 53 | + 20 | + 20 | + 7 | + 28 | + 7 | +13 | 0 |
| 30 | 40 | + 75 | + 50 | +150 | + 50 | +210 | + 50 | +300 | + 50 | + 41 | + 25 | + 50 | + 25 | + 64 | + 25 | + 25 | + 9 | + 34 | + 9 | +16 | 0 |
| 40 | 50 | | | | | | | | | | | | | | | | | | | | |
| 50 | 65 | + 90 | + 60 | +180 | + 60 | +250 | + 60 | +360 | + 60 | + 49 | + 30 | + 60 | + 30 | + 76 | + 30 | + 29 | +10 | + 40 | +10 | +19 | 0 |
| 65 | 80 | | | | | | | | | | | | | | | | | | | | |
| 80 | 100 | +107 | + 72 | +212 | + 72 | +292 | + 72 | +422 | + 72 | + 58 | + 36 | + 71 | + 36 | + 90 | + 36 | + 34 | +12 | + 47 | +12 | +22 | 0 |
| 100 | 120 | | | | | | | | | | | | | | | | | | | | |
| 120 | 140 | +125 | + 85 | +245 | + 85 | +335 | + 85 | +485 | + 85 | + 68 | + 43 | + 83 | + 43 | +106 | + 43 | + 39 | +14 | + 54 | +14 | +25 | 0 |
| 140 | 160 | | | | | | | | | | | | | | | | | | | | |
| 160 | 180 | | | | | | | | | | | | | | | | | | | | |
| 180 | 200 | +146 | +100 | +285 | +100 | +390 | +100 | +560 | +100 | + 79 | + 50 | + 96 | + 50 | +122 | + 50 | + 44 | +15 | + 61 | +15 | +29 | 0 |
| 200 | 225 | | | | | | | | | | | | | | | | | | | | |
| 225 | 250 | | | | | | | | | | | | | | | | | | | | |
| 250 | 280 | +162 | +110 | +320 | +110 | +430 | +110 | +630 | +110 | + 88 | + 56 | +108 | + 56 | +137 | + 56 | + 49 | +17 | + 69 | +17 | +32 | 0 |
| 280 | 315 | | | | | | | | | | | | | | | | | | | | |
| 315 | 355 | +182 | +125 | +355 | +125 | +485 | +125 | +695 | +125 | + 98 | + 62 | +119 | + 62 | +151 | + 62 | + 54 | +18 | + 75 | +18 | +36 | 0 |
| 355 | 400 | | | | | | | | | | | | | | | | | | | | |
| 400 | 450 | +198 | +135 | +385 | +135 | +535 | +135 | +765 | +135 | +108 | + 68 | +131 | + 68 | +165 | + 68 | + 60 | +20 | + 83 | +20 | +40 | 0 |
| 450 | 500 | | | | | | | | | | | | | | | | | | | | |
| 500 | 560 | +215 | +145 | – | – | – | – | – | – | +120 | + 76 | +146 | + 76 | +186 | + 76 | + 66 | +22 | + 92 | +22 | +44 | 0 |
| 560 | 630 | | | | | | | | | | | | | | | | | | | | |
| 630 | 710 | +240 | +160 | – | – | – | – | – | – | +130 | + 80 | +160 | + 80 | +205 | + 80 | + 74 | +24 | +104 | +24 | +50 | 0 |
| 710 | 800 | | | | | | | | | | | | | | | | | | | | |
| 800 | 900 | +260 | +170 | – | – | – | – | – | – | +142 | + 86 | +176 | + 86 | +226 | + 86 | + 82 | +26 | +116 | +26 | +56 | 0 |
| 900 | 1 000 | | | | | | | | | | | | | | | | | | | | |
| 1 000 | 1 120 | +300 | +195 | – | – | – | – | – | – | +164 | + 98 | +203 | + 98 | +263 | + 98 | + 94 | +28 | +133 | +28 | +66 | 0 |
| 1 120 | 1 250 | | | | | | | | | | | | | | | | | | | | |
| 1 250 | 1 400 | +345 | +220 | – | – | – | – | – | – | +188 | +110 | +235 | +110 | +305 | +110 | +108 | +30 | +155 | +30 | +78 | 0 |
| 1 400 | 1 600 | | | | | | | | | | | | | | | | | | | | |
| 1 600 | 1 800 | +390 | +240 | – | – | – | – | – | – | +212 | +120 | +270 | +120 | +350 | +120 | +124 | +32 | +182 | +32 | +92 | 0 |
| 1 800 | 2 000 | | | | | | | | | | | | | | | | | | | | |

Unit : $\mu$m

| Diameter division mm Over | incl. | K6 High | K6 Low | K7 High | K7 Low | M6 High | M6 Low | M7 High | M7 Low | N6 High | N6 Low | N7 High | N7 Low | P6 High | P6 Low | P7 High | P7 Low | R6 High | R6 Low | R7 High | R7 Low |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 6 | +2 | − 6 | + 3 | − 9 | − 1 | − 9 | 0 | − 12 | − 5 | − 13 | − 4 | − 16 | − 9 | − 17 | − 8 | − 20 | − 12 | − 20 | − 11 | − 23 |
| 6 | 10 | +2 | − 7 | + 5 | − 10 | − 3 | − 12 | 0 | − 15 | − 7 | − 16 | − 4 | − 19 | − 12 | − 21 | − 9 | − 24 | − 16 | − 25 | − 13 | − 28 |
| 10 | 18 | +2 | − 9 | + 6 | − 12 | − 4 | − 15 | 0 | − 18 | − 9 | − 20 | − 5 | − 23 | − 15 | − 26 | − 11 | − 29 | − 20 | − 31 | − 16 | − 34 |
| 18 | 30 | +2 | −11 | + 6 | − 15 | − 4 | − 17 | 0 | − 21 | −11 | − 24 | − 7 | − 28 | − 18 | − 31 | − 14 | − 35 | − 24 | − 37 | − 20 | − 41 |
| 30 | 40 | +3 | −13 | + 7 | − 18 | − 4 | − 20 | 0 | − 25 | −12 | − 28 | − 8 | − 33 | − 21 | − 37 | − 17 | − 42 | − 29 | − 42 | − 25 | − 50 |
| 40 | 50 | | | | | | | | | | | | | | | | | | | | |
| 50 | 65 | +4 | −15 | + 9 | − 21 | − 5 | − 24 | 0 | − 30 | −14 | − 33 | − 9 | − 39 | − 26 | − 45 | − 21 | − 51 | − 35 | − 54 | − 30 | − 60 |
| 65 | 80 | | | | | | | | | | | | | | | | | − 37 | − 56 | − 32 | − 62 |
| 80 | 100 | +4 | −18 | +10 | − 25 | − 6 | − 28 | 0 | − 35 | −16 | − 38 | −10 | − 45 | − 30 | − 52 | − 24 | − 59 | − 44 | − 66 | − 38 | − 73 |
| 100 | 120 | | | | | | | | | | | | | | | | | − 47 | − 69 | − 41 | − 76 |
| 120 | 140 | +4 | −21 | +12 | − 28 | − 8 | − 33 | 0 | − 40 | −20 | − 45 | − 12 | − 52 | − 36 | − 61 | − 28 | − 68 | − 56 | − 81 | − 48 | − 88 |
| 140 | 160 | | | | | | | | | | | | | | | | | − 58 | − 83 | − 50 | − 90 |
| 160 | 180 | | | | | | | | | | | | | | | | | − 61 | − 86 | − 53 | − 93 |
| 180 | 200 | +5 | −24 | +13 | − 33 | − 8 | − 37 | 0 | − 46 | −22 | − 51 | −14 | − 60 | − 41 | − 70 | − 33 | − 79 | − 68 | − 97 | − 60 | −106 |
| 200 | 225 | | | | | | | | | | | | | | | | | − 71 | −100 | − 63 | −109 |
| 225 | 250 | | | | | | | | | | | | | | | | | − 75 | −104 | − 67 | −113 |
| 250 | 280 | +5 | −27 | +16 | − 36 | − 9 | − 41 | 0 | − 52 | −25 | − 57 | −14 | − 66 | − 47 | − 79 | − 36 | − 88 | − 85 | −117 | − 74 | −126 |
| 280 | 315 | | | | | | | | | | | | | | | | | − 89 | −121 | − 78 | −130 |
| 315 | 355 | +7 | −29 | +17 | − 40 | −10 | − 46 | 0 | − 57 | −26 | − 62 | −16 | − 73 | − 51 | − 87 | − 41 | − 98 | − 97 | −133 | − 87 | −144 |
| 355 | 400 | | | | | | | | | | | | | | | | | −103 | −139 | − 93 | −150 |
| 400 | 450 | +8 | −32 | +18 | − 45 | −10 | − 50 | 0 | − 63 | −27 | − 67 | −17 | − 80 | − 55 | − 95 | − 45 | −108 | −113 | −153 | −103 | −166 |
| 450 | 500 | | | | | | | | | | | | | | | | | −119 | −159 | −109 | −172 |
| 500 | 560 | 0 | −44 | 0 | − 70 | −26 | − 70 | −26 | − 96 | −44 | − 88 | −44 | −114 | − 78 | −122 | − 78 | −148 | −150 | −194 | −150 | −220 |
| 560 | 630 | | | | | | | | | | | | | | | | | −155 | −199 | −155 | −225 |
| 630 | 710 | 0 | −50 | 0 | − 80 | −30 | − 80 | −30 | −100 | −50 | −100 | −50 | −130 | − 88 | −138 | − 88 | −168 | −175 | −225 | −175 | −225 |
| 710 | 800 | | | | | | | | | | | | | | | | | −185 | −235 | −185 | −265 |
| 800 | 900 | 0 | −56 | 0 | − 90 | −34 | − 90 | −34 | − 124 | −56 | −112 | −56 | −146 | −100 | −156 | −100 | −190 | −210 | −266 | −210 | −300 |
| 900 | 1 000 | | | | | | | | | | | | | | | | | −220 | −276 | −220 | −310 |
| 1 000 | 1 120 | 0 | −66 | 0 | −105 | −40 | −106 | −40 | −145 | −66 | −132 | −66 | −171 | −120 | −186 | −120 | −225 | −250 | −316 | −250 | −355 |
| 1 120 | 1 250 | | | | | | | | | | | | | | | | | −260 | −326 | −260 | −365 |
| 1 250 | 1 400 | 0 | −78 | 0 | −125 | −48 | −126 | −48 | −173 | −78 | −156 | −78 | −203 | −140 | −213 | −140 | −265 | −300 | −378 | −300 | −425 |
| 1 400 | 1 600 | | | | | | | | | | | | | | | | | −330 | −408 | −330 | −455 |
| 1 600 | 1 800 | 0 | −92 | 0 | −150 | −58 | −150 | −58 | −208 | −92 | −184 | −92 | −242 | −170 | −262 | −170 | −320 | −370 | −462 | −370 | −520 |
| 1 800 | 2 000 | | | | | | | | | | | | | | | | | −400 | −492 | −400 | −550 |

Unit : μm

| H7 | | H8 | | H9 | | H10 | | H11 | | H13 | | J6 | | Js6 | | J7 | | Js7 | | K5 | | Diameter division mm | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | High | Low | Over | incl. |
| + 12 | 0 | + 18 | 0 | + 30 | 0 | + 48 | 0 | + 75 | 0 | +180 | 0 | + 5 | −3 | + 4 | − 4 | + 6 | − 6 | + 6 | − 6 | 0 | − 5 | 3 | 6 |
| + 15 | 0 | + 22 | 0 | + 36 | 0 | + 58 | 0 | + 90 | 0 | +220 | 0 | + 5 | −4 | + 4.5 | − 4.5 | + 8 | − 7 | + 7.5 | − 7.5 | +1 | − 5 | 6 | 10 |
| + 18 | 0 | + 27 | 0 | + 43 | 0 | + 70 | 0 | +110 | 0 | +270 | 0 | + 6 | −5 | + 5.5 | − 5.5 | +10 | − 8 | + 9 | − 9 | +2 | − 6 | 10 | 18 |
| + 21 | 0 | + 33 | 0 | + 52 | 0 | + 84 | 0 | +130 | 0 | +330 | 0 | + 8 | −5 | + 6.5 | − 6.5 | +12 | − 9 | +10.5 | −10.5 | +1 | − 8 | 18 | 30 |
| + 25 | 0 | + 39 | 0 | + 62 | 0 | +100 | 0 | +160 | 0 | +390 | 0 | +10 | −6 | + 8 | − 8 | +14 | −11 | +12.5 | −12.5 | +2 | − 9 | 30<br>40 | 40<br>50 |
| + 30 | 0 | + 46 | 0 | + 74 | 0 | +120 | 0 | +190 | 0 | +460 | 0 | +13 | −6 | + 9.5 | − 9.5 | +18 | −12 | +15 | −15 | +3 | −10 | 50<br>65 | 65<br>80 |
| + 35 | 0 | + 54 | 0 | + 87 | 0 | +140 | 0 | +220 | 0 | +540 | 0 | +16 | −6 | +11 | −11 | +22 | −13 | +17.5 | −17.5 | +2 | −13 | 80<br>100 | 100<br>120 |
| + 40 | 0 | + 63 | 0 | +100 | 0 | +160 | 0 | +250 | 0 | +630 | 0 | +18 | −7 | +12.5 | −12.5 | +26 | −14 | +20 | −20 | +3 | −15 | 120<br>140<br>160 | 140<br>160<br>180 |
| + 46 | 0 | + 72 | 0 | +115 | 0 | +185 | 0 | +290 | 0 | +720 | 0 | +22 | −7 | +14.5 | −14.5 | +30 | −16 | +23 | −23 | +2 | −18 | 180<br>200<br>225 | 200<br>225<br>250 |
| + 52 | 0 | + 81 | 0 | +130 | 0 | +210 | 0 | +320 | 0 | +810 | 0 | +25 | −7 | +16 | −16 | +36 | −16 | +26 | −26 | +3 | −20 | 250<br>280 | 280<br>315 |
| + 57 | 0 | + 89 | 0 | +140 | 0 | +230 | 0 | +360 | 0 | +890 | 0 | +29 | −7 | +18 | −18 | +39 | −18 | +28.5 | −28.5 | +3 | −22 | 315<br>355 | 355<br>400 |
| + 63 | 0 | + 97 | 0 | +155 | 0 | +250 | 0 | +400 | 0 | +970 | 0 | +33 | −7 | +20 | −20 | +43 | −20 | +31.5 | −31.5 | +2 | −25 | 400<br>450 | 450<br>500 |
| + 70 | 0 | +110 | 0 | +175 | 0 | +280 | 0 | +440 | 0 | − | 0 | − | − | +22 | −22 | − | − | +35 | −35 | − | − | 500<br>560 | 560<br>630 |
| + 80 | 0 | +125 | 0 | +200 | 0 | +320 | 0 | +500 | 0 | − | 0 | − | − | +25 | −25 | − | − | +40 | −40 | − | − | 630<br>710 | 710<br>800 |
| + 90 | 0 | +140 | 0 | +230 | 0 | +360 | 0 | +560 | 0 | − | 0 | − | − | +28 | −28 | − | − | +45 | −45 | − | − | 800<br>900 | 900<br>1 000 |
| +105 | 0 | +165 | 0 | +260 | 0 | +420 | 0 | +660 | 0 | − | 0 | − | − | +33 | −33 | − | − | +52.5 | −52.5 | − | − | 1 000<br>1 120 | 1 120<br>1 250 |
| +125 | 0 | +195 | 0 | +310 | 0 | +500 | 0 | +780 | 0 | − | 0 | − | − | +39 | −39 | − | − | +62.5 | −62.5 | − | − | 1 250<br>1 400 | 1 400<br>1 600 |
| +150 | 0 | +230 | 0 | +370 | 0 | +600 | 0 | +920 | 0 | − | 0 | − | − | +46 | −46 | − | − | +75 | −75 | − | − | 1 600<br>1 800 | 1 800<br>2 000 |

Inch - Millimeter conversion table

1in.=25.4mm

| Inch | | 0" | 1" | 2" | 3" | 4" | 5" | 6" | 7" | 8" | 9" |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Fraction | Decimals | | | | | | | | | | |
| | | | 25.400 | 50.800 | 76.200 | 101.600 | 127.000 | 152.400 | 177.800 | 203.200 | 228.600 |
| 1/64 | 0.015625 | 0.397 | 25.797 | 51.197 | 76.597 | 101.997 | 127.397 | 152.797 | 178.197 | 203.597 | 228.997 |
| 1/32 | 0.031250 | 0.794 | 26.194 | 51.594 | 76.994 | 102.394 | 127.794 | 153.194 | 178.594 | 203.994 | 229.394 |
| 3/64 | 0.046875 | 1.191 | 26.591 | 51.991 | 77.391 | 102.791 | 128.191 | 153.591 | 178.991 | 204.391 | 229.791 |
| 1/16 | 0.062500 | 1.588 | 26.988 | 52.388 | 77.788 | 103.188 | 128.588 | 153.988 | 179.388 | 204.788 | 230.188 |
| 5/64 | 0.078125 | 1.984 | 27.384 | 52.784 | 78.184 | 103.584 | 128.984 | 154.384 | 179.784 | 205.184 | 230.584 |
| 3/32 | 0.093750 | 2.381 | 27.781 | 53.181 | 48.581 | 103.981 | 129.381 | 154.781 | 180.181 | 205.581 | 230.981 |
| 7/64 | 0.109375 | 2.778 | 28.178 | 53.578 | 78.978 | 104.378 | 129.778 | 155.178 | 180.578 | 205.978 | 231.378 |
| 1/ 8 | 0.125000 | 3.175 | 28.575 | 53.975 | 79.375 | 104.775 | 130.175 | 155.575 | 180.975 | 206.375 | 231.775 |
| 9/64 | 0.140625 | 3.572 | 28.972 | 54.372 | 79.772 | 105.172 | 130.572 | 155.972 | 181.372 | 206.772 | 232.172 |
| 5/32 | 0.156250 | 3.969 | 29.369 | 54.769 | 80.169 | 105.569 | 130.969 | 156.369 | 181.769 | 207.169 | 232.569 |
| 11/64 | 0.171875 | 4.366 | 29.766 | 55.166 | 80.566 | 105.966 | 131.366 | 156.766 | 182.166 | 207.566 | 232.966 |
| 3/16 | 0.187500 | 4.762 | 30.162 | 55.562 | 80.962 | 106.362 | 131.762 | 157.162 | 182.562 | 207.962 | 233.362 |
| 13/64 | 0.203125 | 5.159 | 30.559 | 55.959 | 81.359 | 106.759 | 132.159 | 157.559 | 182.959 | 208.359 | 233.759 |
| 7/32 | 0.218750 | 5.556 | 30.956 | 56.356 | 81.756 | 107.156 | 132.556 | 157.956 | 183.356 | 208.756 | 234.156 |
| 15/64 | 0.234375 | 5.953 | 31.353 | 56.753 | 82.153 | 107.553 | 132.953 | 158.353 | 183.753 | 209.153 | 234.553 |
| 1/ 4 | 0.250000 | 6.350 | 31.750 | 57.150 | 82.550 | 107.950 | 133.350 | 158.750 | 184.150 | 209.550 | 234.950 |
| 17/64 | 0.265625 | 6.747 | 32.147 | 57.547 | 82.947 | 108.347 | 133.747 | 159.147 | 184.547 | 209.947 | 235.347 |
| 9/32 | 0.281250 | 7.144 | 32.544 | 57.944 | 83.344 | 108.744 | 134.144 | 159.544 | 184.944 | 210.344 | 235.744 |
| 19/64 | 0.296875 | 7.541 | 32.941 | 58.341 | 83.741 | 109.141 | 134.541 | 159.941 | 185.341 | 210.741 | 236.141 |
| 5/16 | 0.312500 | 7.938 | 33.338 | 58.738 | 84.138 | 109.538 | 134.938 | 160.338 | 185.738 | 211.138 | 236.538 |
| 21/64 | 0.328125 | 8.334 | 33.734 | 59.134 | 84.534 | 109.934 | 135.334 | 160.734 | 186.134 | 211.534 | 236.934 |
| 11/32 | 0.343750 | 8.731 | 34.131 | 59.531 | 84.931 | 110.331 | 135.731 | 161.131 | 186.531 | 211.931 | 237.331 |
| 23/64 | 0.359375 | 9.128 | 34.528 | 59.928 | 85.328 | 110.728 | 136.128 | 161.528 | 186.928 | 212.328 | 237.728 |
| 3/ 8 | 0.375000 | 9.525 | 34.925 | 60.325 | 85.725 | 111.125 | 136.525 | 161.925 | 187.325 | 212.725 | 238.125 |
| 25/64 | 0.390625 | 9.922 | 35.322 | 60.722 | 86.122 | 111.522 | 136.922 | 162.322 | 187.722 | 213.122 | 238.522 |
| 13/32 | 0.406250 | 10.319 | 35.719 | 61.119 | 86.519 | 111.919 | 137.319 | 162.719 | 188.119 | 213.519 | 238.919 |
| 27/64 | 0.421875 | 10.716 | 36.116 | 61.516 | 86.916 | 112.316 | 137.716 | 163.116 | 188.516 | 213.916 | 239.316 |
| 7/16 | 0.437500 | 11.112 | 36.512 | 61.912 | 87.312 | 112.721 | 138.112 | 163.512 | 188.912 | 214.312 | 239.712 |
| 29/64 | 0.453125 | 11.509 | 36.909 | 62.309 | 87.709 | 113.109 | 138.509 | 163.909 | 189.309 | 214.709 | 240.109 |
| 15/32 | 0.468750 | 11.906 | 37.306 | 62.706 | 88.106 | 113.506 | 138.906 | 164.306 | 189.706 | 215.106 | 240.506 |
| 31/64 | 0.484375 | 12.303 | 37.703 | 63.103 | 88.503 | 113.903 | 139.303 | 164.703 | 190.103 | 215.503 | 240.903 |
| 1/ 2 | 0.500000 | 12.700 | 38.100 | 63.500 | 88.900 | 114.300 | 139.700 | 165.100 | 190.500 | 215.900 | 241.300 |
| 33/64 | 0.515625 | 13.097 | 38.497 | 63.897 | 89.297 | 114.697 | 140.097 | 165.497 | 190.897 | 216.297 | 241.697 |
| 17/32 | 0.531250 | 13.494 | 38.894 | 64.294 | 89.694 | 115.094 | 140.494 | 165.894 | 191.294 | 216.694 | 242.094 |
| 35/64 | 0.546875 | 13.891 | 39.291 | 64.691 | 90.091 | 115.491 | 140.891 | 166.291 | 191.691 | 217.091 | 242.491 |
| 9/16 | 0.562500 | 14.288 | 39.688 | 65.088 | 90.488 | 115.888 | 141.283 | 166.688 | 192.088 | 217.488 | 242.888 |
| 37/64 | 0.578125 | 14.684 | 40.084 | 65.484 | 90.884 | 116.284 | 141.684 | 167.084 | 192.484 | 217.884 | 243.284 |
| 19/32 | 0.593750 | 15.081 | 40.481 | 65.881 | 91.281 | 116.681 | 142.081 | 167.481 | 192.881 | 218.281 | 243.681 |
| 39/64 | 0.609375 | 15.478 | 40.878 | 66.278 | 91.678 | 117.078 | 142.478 | 167.878 | 193.278 | 218.678 | 244.078 |
| 5/ 8 | 0.625000 | 15.875 | 41.275 | 66.675 | 92.075 | 117.475 | 142.875 | 168.275 | 193.675 | 219.075 | 244.475 |
| 41/64 | 0.640625 | 16.272 | 41.672 | 67.072 | 92.472 | 117.872 | 143.272 | 168.672 | 194.072 | 219.472 | 244.872 |
| 21/32 | 0.656250 | 16.669 | 42.069 | 67.469 | 92.869 | 118.269 | 143.669 | 169.069 | 194.469 | 219.869 | 245.269 |
| 43/64 | 0.671875 | 17.066 | 42.466 | 67.866 | 93.266 | 118.666 | 144.066 | 169.466 | 194.866 | 220.266 | 245.666 |
| 11/16 | 0.687500 | 17.462 | 42.862 | 68.262 | 93.662 | 119.062 | 144.462 | 169.862 | 195.262 | 220.662 | 246.062 |
| 45/64 | 0.703125 | 17.859 | 43.259 | 68.659 | 94.059 | 119.459 | 144.859 | 170.259 | 195.659 | 221.056 | 246.459 |
| 23/32 | 0.718750 | 18.256 | 43.656 | 69.056 | 94.456 | 119.856 | 145.256 | 170.656 | 196.056 | 221.456 | 246.856 |
| 47/64 | 0.734375 | 18.653 | 44.053 | 69.453 | 94.853 | 120.253 | 145.653 | 171.053 | 196.453 | 221.853 | 247.253 |
| 3/ 4 | 0.750000 | 19.050 | 44.450 | 69.850 | 95.250 | 120.650 | 146.050 | 171.450 | 196.850 | 222.250 | 247.650 |
| 49/64 | 0.765625 | 19.447 | 44.847 | 70.247 | 95.647 | 121.047 | 146.447 | 171.847 | 197.247 | 222.647 | 248.047 |
| 25/32 | 0.781250 | 19.844 | 45.244 | 70.644 | 96.044 | 121.444 | 146.844 | 172.244 | 197.644 | 223.044 | 248.444 |
| 51/64 | 0.796875 | 20.241 | 45.641 | 71.041 | 96.441 | 121.841 | 147.241 | 172.641 | 198.041 | 223.441 | 248.841 |
| 13/16 | 0.812500 | 20.638 | 46.038 | 71.438 | 96.838 | 122.238 | 147.638 | 173.038 | 198.438 | 223.838 | 249.238 |
| 53/64 | 0.828125 | 21.034 | 46.434 | 71.834 | 97.234 | 122.634 | 148.034 | 173.434 | 198.834 | 224.234 | 249.634 |
| 27/32 | 0.843750 | 21.431 | 46.831 | 72.231 | 97.631 | 123.031 | 148.431 | 173.831 | 199.231 | 224.631 | 250.031 |
| 55/64 | 0.859375 | 21.828 | 47.228 | 72.628 | 98.028 | 123.428 | 148.828 | 174.228 | 199.628 | 225.028 | 250.428 |
| 7/ 8 | 0.875000 | 22.225 | 47.625 | 73.025 | 98.425 | 123.825 | 149.225 | 174.625 | 200.025 | 225.425 | 250.825 |
| 57/64 | 0.890625 | 22.622 | 48.022 | 73.422 | 98.822 | 124.222 | 149.622 | 175.022 | 200.422 | 225.822 | 251.222 |
| 39/32 | 0.906250 | 23.019 | 48.419 | 73.819 | 99.219 | 124.619 | 150.019 | 175.419 | 200.819 | 226.219 | 251.619 |
| 59/64 | 0.921875 | 23.416 | 48.816 | 74.216 | 99.616 | 125.016 | 150.416 | 175.816 | 201.216 | 226.616 | 252.016 |
| 15/16 | 0.937500 | 23.812 | 49.212 | 74.612 | 100.012 | 125.412 | 150.812 | 176.212 | 201.612 | 227.012 | 252.412 |
| 61/64 | 0.953125 | 24.209 | 49.609 | 75.009 | 100.409 | 125.809 | 151.209 | 176.609 | 202.009 | 227.409 | 252.809 |
| 31/32 | 0.968750 | 24.606 | 50.006 | 75406 | 100.806 | 126.206 | 151.606 | 177.006 | 202.406 | 227.806 | 253.206 |
| 63/64 | 0.984375 | 25.003 | 50.403 | 75.803 | 101.203 | 126.603 | 152.003 | 177.403 | 202.803 | 228.203 | 253.603 |